Manifold-Aware Concept Erasure

Anonymous Author(s)
Affiliation
Address
email

Abstract

Concept erasure aims to remove target information from a representation without destroying other information it carries. This is difficult for nonlinear concepts: removing the signal used by one probe does not prevent a fresh nonlinear probe from recovering the concept, while unconstrained gradient-based edits can move representations away from the support of natural inputs. We introduce TanGCE, a manifold-aware erasure method that constrains nonlinear edits to local tangent directions. For each representation, TanGCE estimates a tangent basis from nearest-neighbor secants, projects the concept-scorer gradient onto this basis, and scales the resulting step by the local singular spectrum, so motion along each tangent axis is proportional to the neighborhood's extent along that axis. Before the iterative tangent loop, we remove dominant low-order target signal with closed-form LEACE and a rank-2 covariance-asymmetry projection. Across 119 panels, covering 13 language models, three NLP concepts, and 40 CelebA-CLIP attributes under two control regimes, TanGCE uses one fixed set of hyperparameters with no per-panel tuning. On the 39 NLP panels, it reaches the majority-class probe-accuracy floor on 26–33 panels across control-damage budgets, compared with 22–26 for Obliviator, while preserving control accuracy better. On CelebA-CLIP, it is the only method that retains an admissible operating point for every attribute in the highly correlated control regime.

1 Introduction

Neural representations encode many attributes of their inputs: semantic content, style, demographic information, task labels, and other latent factors. In many applications, we may want to remove one such attribute, for example a protected or spurious concept, while preserving everything else the representation contains [De-Arteaga et al., 2019, Avitan et al., 2025]. This is the central tension in *concept erasure*: removing the target information while unnecessarily damaging other control concepts, known as *surgicality* (Yanai:) is it actually known? have previous papers used this term? mayes!.

The surgicality requirement makes erasure especially difficult. A target concept is rarely encoded in isolation. It may be entangled with many other factors, most of which are unlabeled and therefore cannot be explicitly protected. Moreover, erasure must address *recoverability*: even when a chosen classifier is driven to chance, a freshly trained classifier on the edited representations may still recover the protected attribute [Elazar and Goldberg, 2018]. Linear erasure methods provide simple and effective closed-form interventions [Ravfogel et al., 2020, Belrose et al., 2023], but they only remove linearly accessible information and do not generally block nonlinear probes [Ravfogel et al., 2022]. Nonlinear erasure, however, introduces a different danger: if we directly follow the gradient of a nonlinear concept scorer in the ambient representation space, the edit may remove target signal by moving the representation to a region that natural inputs never occupy.

While neural representations live in $\mathbb{R}^d$, learned representations that are produced by different models do not fill this space uniformly. Instead, they concentrate near a much smaller, structured support. This observation changes the geometry of erasure; an arbitrary direction in the ambient space may be a valid vector in $\mathbb{R}^d$, but it need not be a valid direction along the representation support. Moving in

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

such a direction can push the representation off-support, distorting unrelated information and harming control concepts. We therefore ask whether nonlinear (Yanai:) why is this related to non linearity? erasure can be made more surgical by forcing the edit to remain locally compatible with the geometry of natural representations.

We introduce TanGCE, a tangent gradient-based method for nonlinear concept erasure. Around each representation, we estimate a local tangent space using the SVD of nearest-neighbor secants. Instead of applying the ambient gradient (Yanai:) what's that? of a concept scorer, we project that gradient onto the estimated tangent space and edit only along directions supported by the local neighborhood. This constrains erasure to directions that are locally plausible under the *representation manifold*, reducing the risk of off-support interventions.

A second ingredient is needed for stability. The local tangent space is not isotropic: different tangent directions correspond to different local scales of variation. A step of the same Euclidean length can be harmless along a high-variance direction but destructive along a low-variance one. We therefore use the singular spectrum produced by the same local SVD to precondition the tangent step. After projecting the scorer gradient onto the tangent space, we express it in the local principal-coordinate system and rescale each coordinate by its corresponding local scale. This local-σ preconditioner ties the magnitude of the intervention in each direction to how far the representation neighborhood actually extends in that direction, yielding a per-row trust region that transfers across models and layers.

Our full pipeline has two stages. First, we remove dominant low-order target signal in closed form. We apply LEACE to remove the first-order mean-difference direction between the two target classes [Belrose et al., 2023], and then remove a rank-2 subspace corresponding to the leading covariance-asymmetry directions of the class-conditional covariance difference. This sacrifices only a small number of representation dimensions while reducing the strongest linear and quadratic traces of the target concept. Second, we run tangent-constrained nonlinear erasure on the remaining signal using the local-σ preconditioned tangent gradients described above.

(Yanai:) the entire method is described across three paragraphs - that's too much. make it more concise

The empirical signature of this design is surgicality (Yanai:) strange phrasing. We evaluate TanGCE on 119 panels (Yanai:) idk what panels are (well, you told me, but that's not a common term). experiments/setups? under fixed pipeline hyperparameters, without per-panel tuning: 13 language models $\times$ 3 concepts at 50%-depth, and CelebA-CLIP $\times$ 40 attributes $\times$ 2 control-set regimes. At a fixed control-damage budget of $\Delta Y \geq -5$pp, TanGCE reaches the per-panel majority-class probe-accuracy floor on 33 of 39 NLP panels, compared with $22/39$ for Obliviator, while damaging the control task less; the same pattern holds at every other budget we test ($\leq$ 1pp, $\leq$ 3pp, $\leq$ 10pp). (Yanai:) you use many terms here w/o explaining them (panel, control-set regimes, damage budget, NLP panel, Obliviator). re-write this paragraph. focus on the high level idea: we remove stuff, we measure both how much was removed, and how much it affected other concepts. we test the method on text and vision tasks, X models, Y tasks, and show preferable results over the baselines

Our contributions are:

1. We formulate nonlinear concept erasure as a geometry-constrained intervention problem: target information should be removed without moving representations away from the local support of natural inputs.
2. We introduce TanGCE, a two-stage erasure pipeline combining closed-form low-order preprocessing with tangent-constrained nonlinear erasure.
3. We propose a local-σ preconditioner that scales tangent-coordinate steps by the local singular spectrum, making the trust region sensitive to the manifold's anisotropic geometry.
4. We evaluate TanGCE on 119 panels under a single fixed-ε protocol, showing stronger target removal and control preservation than Obliviator across NLP and vision-language settings.

2 Related Work

Concept erasure methods differ mainly in the function class against which they erase and in the geometry they assume. Linear methods such as INLP [Ravfogel et al., 2020] and LEACE [Belrose

et al., 2023] target linearly decodable concept information. INLP removes successively exposed linear directions by iterative nullspace projection, while LEACE gives the strongest closed-form linear baseline in our comparisons. These methods are essential reference points, but they do not directly address the case where a fresh nonlinear probe can still recover the concept after linear erasure.

That recoverability concern is central to our framing. Elazar and Goldberg [Elazar and Goldberg, 2018] showed that apparent success against one learned remover or classifier does not imply that the protected information is gone for a newly trained downstream model. Amnesic probing [Elazar et al., 2021] and related intervention-based analyses adopt the same conservative lesson: the meaningful question is what remains recoverable after the intervention, not only what the original scorer no longer captures.

Among nonlinear erasure methods, IGBP [Iskander et al., 2023] is the closest prior baseline because it iteratively trains a nonlinear probe and retracts representations toward the current decision boundary. We report INLP and IGBP under the same trajectory-probe protocol as every other method in Tab. 1. Our paper differs in two ways. First, AmbGCE is introduced only as the direct ambient-space ablation that keeps the iterative nonlinear scorer loop but removes the manifold constraint. Second, TanGCE is the paper's main method: it projects the erasure update onto a locally estimated tangent space, making the geometric hypothesis itself the main point of comparison.

3 Problem setup and geometric motivation

Let $\mathrm{enc}_\ell : \Sigma^* \to \mathbb{R}^d$ be the hidden state at layer ℓ of a language model over vocabulary Σ. For each sample i we have a representation $\mathbf{x}_i = \mathrm{enc}_\ell(s_i) \in \mathbb{R}^d$, a binary concept label $y_i \in \{0, 1\}$, and an optional control label z_i. The dataset is $\mathcal{D} = \{(\mathbf{x}_i, y_i, z_i)\}_{i=1}^N$ with class subsets $\mathcal{D}_c = \{i : y_i = c\}$. A concept scorer $f_\theta : \mathbb{R}^d \to \mathbb{R}$ is a 2-layer MLP with input gradient $\nabla f_\theta(\mathbf{x}_i)$.

Local manifold hypothesis. We adopt the modeling assumption that the representations $\{\mathbf{x}_i\}$ lie near a locally low-dimensional support $\mathcal{M} \subset \mathbb{R}^d$. Its operational consequence is that we can approximate the neighborhood of any $\mathbf{x} \in \mathcal{M}$ by its tangent space $T_\mathbf{x}\mathcal{M}$ [Tu, 2011] and apply linear-algebraic operations (SVD, projection, anisotropic scaling) inside it. The tangent space is not observed directly; we estimate it from nearby representations by local SVD on the secant matrix (§4). For an ambient vector $\mathbf{g} \in \mathbb{R}^d$, the decomposition $\mathbf{g} = \mathbf{g}^\top + \mathbf{g}^\perp$ with $\mathbf{g}^\top \in T_\mathbf{x}\mathcal{M}$ separates the part of $\mathbf{g}$ supported by the local sample structure from the part orthogonal to it. TanGCE acts on the tangent component only; AmbGCE retains both. Formal coordinate-chart definitions and the differential $d(\varphi^{-1})_0$ identification with $T_\mathbf{x}\mathcal{M}$ are deferred to App. I.

Recoverability. Concept removal is judged only after training a *fresh* probe on the edited representations [Elazar and Goldberg, 2018]: erasure succeeds if a newly trained nonlinear classifier fails to recover the concept from $\tilde{\mathbf{x}}$, not merely if the old scorer is fooled.

4 Method: TanGCE

TanGCE (*Tangent Gradient-based Concept Erasure*) has two phases. Phase 1 applies two closed-form moment-matching projections that exhaust 1st- and 2nd-moment concept signal. Phase 2 iterates a tangent-constrained step on the residual: each round trains a fresh nonlinear scorer, projects its gradient onto a locally estimated tangent basis, and applies the projection-and-preconditioning update.

4.1 Phase 1: closed-form moment preprocessing

LEACE (1st moment). Fit a LEACE eraser [Belrose et al., 2023] on $(\mathbf{X}_{\text{train}}, \mathbf{y})$ and apply to all splits. LEACE removes the rank-1 cross-covariance $\mathbf{\Sigma}_{xz}$, leaving no *linear* classifier above majority-class accuracy with provably minimum representation damage among affine projections.

Covariance-asymmetry projection (2nd moment). LEACE matches class means but leaves the within-class asymmetry

$$\Delta\mathbf{\Sigma} = \mathbf{\Sigma}_+ - \mathbf{\Sigma}_-. \tag{1}$$

Algorithm 1 TanGCE at a glance. Phase 1: LEACE + rank-2 covariance-asymmetry projection. Phase 2: iterative tangent-constrained step.

Require: Representations $\mathbf{X}$, concept labels $\mathbf{y}$, #rounds T, trust region ε, anisotropy α, scorer hidden h.

Phase 1 – closed-form moment preprocessing (Eqs. 1–3):

  1a. $\mathbf{X} \leftarrow \mathrm{LEACE}_{\mathbf{y}}(\mathbf{X})$ (rank-1 mean projection)

  1b. $\mathbf{X} \leftarrow \mathbf{X} - \mathbf{D}\mathbf{D}^\top \mathbf{X}$, with $\mathbf{D}$ = top-2 eigvecs of $\Delta\mathbf{\Sigma}$ (rank-2 covariance projection)

Phase 2 – iterative tangent step (Eqs. 6–12):

for $t = 1, \ldots, T$ **do**

  Train a fresh 2-layer MLP scorer f_t (hidden h) on the current $(\mathbf{X}, \mathbf{y})$.

  For each row $\mathbf{x}_i$: tangent-projected, σ-preconditioned trust-region step

$$\mathbf{x}_i \;\leftarrow\; \mathbf{x}_i - \lambda_i \sum_{k=1}^{r} \sigma_{i,k}^{\alpha}\, a_{i,k}\, \mathbf{v}_i^{(k)}, \quad \|\Delta\mathbf{x}_i\|_2 \le \varepsilon \cdot s_i, \quad \alpha = 1.$$

end for

return edited $\mathbf{X}$.

A nonlinear probe still recovers concept information from variance asymmetry along axes where $\mathbf{\Sigma}_+$ and $\mathbf{\Sigma}_-$ differ. We append a rank-2 projection: the top-2 eigenvectors of $\Delta\mathbf{\Sigma}$ ranked by $|\lambda|$, orthonormalized with a residual mean direction via QR,

$$\mathbf{D} = \mathrm{QR}\Big(\big[\,\hat{\mathbf{d}}_{\text{mean}} \;\big|\; \mathbf{e}_1, \mathbf{e}_2\,\big]\Big) \in \mathbb{R}^{d\times 3}, \tag{2}$$

$$\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{D}\mathbf{D}^\top \mathbf{x}. \tag{3}$$

Because LEACE has aligned class means, $\hat{\mathbf{d}}_{\text{mean}} \approx \mathbf{0}$ and the effective rank removed is two. This is a special case of the k-LEACE framework of Singh et al. [2024] restricted to second-order eigenvectors, which are stable to estimate from $n \approx 7000$ examples while higher-order tensors are not. We fix $K{=}2$ across all 119 panels: for a binary concept, the top-2 eigenvectors of $\Delta\mathbf{\Sigma}$ carry the two axes of class-conditional variance asymmetry; subsequent eigenvectors describe noise rather than class-discriminative signal. Empirically, $K{\ge}3$ adds no leakage reduction and begins to erode control accuracy (see Tab. 4).

The two stages erase structurally distinct signal (mean shift vs. variance asymmetry) and together remove rank ≤ 3 from $d \in \{2048, 3072, 3584\}$, exposing the residual nonlinear structure that Phase 2 attacks.

4.2 Local tangent estimation

For each query $\mathbf{x}_i$, let $\mathcal{N}_k(\mathbf{x}_i)$ be its k nearest neighbors. Stack the centered secants $\mathbf{s}_{j_m} = \mathbf{x}_{j_m} - \mathbf{x}_i$ row-wise into $\mathbf{S}_i \in \mathbb{R}^{k\times d}$ and take its SVD,

$$\mathbf{S}_i = \mathbf{U}_i \,\mathrm{diag}(\sigma_{i,1}, \ldots, \sigma_{i,k})\, \mathbf{V}_i^\top. \tag{4}$$

The right singular vectors $\mathbf{V}_i$ are the directions in which neighbors extend; the singular values $\sigma_{i,k}$ are how far they extend along each. The implied local covariance is $\mathbf{C}_i = \frac{1}{k}\mathbf{V}_i \,\mathrm{diag}(\sigma_{i,k}^2)\, \mathbf{V}_i^\top$. Truncating to the top r right singular vectors gives the tangent basis

$$\mathbf{B}_i = [\mathbf{v}_i^{(1)}, \ldots, \mathbf{v}_i^{(r)}] \in \mathbb{R}^{d\times r}, \tag{5}$$

with r estimated per panel by TwoNN [Facco et al., 2017] and floored at $r \ge 8$. The tangent projector is

$$\mathbf{P}_{T_i} = \mathbf{B}_i \mathbf{B}_i^\top. \tag{6}$$

We treat the SVD basis as a first-order stand-in for $T_{\mathbf{x}_i}\mathcal{M}$, not as a guarantee of smooth-manifold structure.

4.3 Tangent-constrained update

The ambient unit gradient at $\mathbf{x}_i$ is $\mathbf{u}_i = \nabla f_t(\mathbf{x}_i)/\|\nabla f_t(\mathbf{x}_i)\|$. TanGCE replaces it with its tangent projection,

$$\mathbf{u}_i^{\tan} = \mathbf{P}_{T_i}\mathbf{u}_i = \mathbf{B}_i(\mathbf{B}_i^\top \mathbf{u}_i), \tag{7}$$

not re-normalized: $\|\mathbf{u}_i^{\tan}\| \leq 1$ acts as an overlap weight that attenuates the update when the ambient gradient has little support in the estimated tangent space. The isotropic update is

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i - \lambda \left(\mathbf{x}_i^\top \mathbf{u}_i^{\tan}\right) \mathbf{u}_i^{\tan}, \tag{8}$$

with λ a step-size scalar (replaced below by a closed-form per-sample λ_i). AmbGCE is the direct ablation: same loop, same scorer, same λ, but applies the update in the full ambient space without the projection. We retain Eq. 8 as the conceptual entry point and as the $\alpha = 0$ ablation; the canonical TanGCE update is the σ-preconditioned generalization in §4.5.

4.4 Adaptive per-sample step size: closed-form λ_i

A single global λ over-edits some samples while under-editing others. We replace it with a per-sample step size that respects a *local trust region*: no sample moves more than a fraction ε of a per-sample local scale s_i,

$$\|\tilde{\mathbf{x}}_i - \mathbf{x}_i\|_2 \leq \varepsilon \cdot s_i. \tag{9}$$

Substituting Eq. 8 and letting $\mathbf{v}_i = (\mathbf{x}_i^\top \mathbf{u}_i^{\tan})\mathbf{u}_i^{\tan}$ be the per-sample displacement at λ=1, the largest admissible step is

$$\boxed{\lambda_i = \min\left(\lambda_{\max},\ \frac{\varepsilon \cdot s_i}{\|\mathbf{v}_i\|}\right)} \tag{10}$$

with $\lambda_{\max} = 64$ a safety cap when $\|\mathbf{v}_i\| \to 0$. We use $\varepsilon = 0.1$ across every panel.

Anisotropy-aware local scale (FMaHa). The local manifold is an ellipsoid: along high-σ singular directions, neighbors extend much further than along thin tail directions. We therefore use the *whitened-tangent neighborhood radius*

$$s_i = R_i^{\text{white}} \cdot s_i^{\text{maha}}, \qquad R_i^{\text{white}} = \text{median}_{j \in \mathcal{N}_k(i)} \|\mathbf{B}_i^\top (\mathbf{x}_j - \mathbf{x}_i)/\boldsymbol{\sigma}_i\|_2, \tag{11}$$

with $\boldsymbol{\sigma}_i = (\sigma_{i,1}, \ldots, \sigma_{i,r})$ and $s_i^{\text{maha}} = 1/\|\boldsymbol{\alpha}_i/(\boldsymbol{\sigma}_i + \delta)\|_2$ (component-wise). ε is then a fraction of the local whitened-tangent neighborhood; setting $s_i = r_i$ (mean k-NN distance) recovers the Euclidean radial scale and is reported in App. as an ablation.

4.5 Anisotropic update via the local σ-spectrum

The closed-form λ_i controls the magnitude of each step, but the direction of Eq. 8 is still isotropic in the tangent basis. The local SVD already exposes anisotropy: the singular values $\sigma_{i,k}$ can vary substantially across tangent axes. An isotropic tangent step can therefore over-edit thin directions where the local linear approximation is fragile, while under-using directions along which the neighborhood actually extends.

We replace Eq. 8 with a local-σ-preconditioned tangent step,

$$\boxed{\tilde{\mathbf{x}}_i \ = \ \mathbf{x}_i \ - \ \lambda_i \sum_{k=1}^{r} \sigma_{i,k}^{\alpha}\, a_{i,k}\, \mathbf{v}_i^{(k)}} \tag{12}$$

where $a_{i,k} = \langle \mathbf{v}_i, \mathbf{v}_i^{(k)} \rangle$ are the coordinates of the unpreconditioned tangent displacement $\mathbf{v}_i$ in the local SVD basis $\{\mathbf{v}_i^{(k)}\}_{k=1}^r$. We use $\alpha = 1$ in the main experiments. Each tangent coefficient is then scaled by the local extent of the neighborhood along that axis: high-σ directions receive larger updates, while thin directions are suppressed.

Relation to local covariance and Mahalanobis scaling. The update can also be viewed as applying a positive spectral power of the local secant covariance. The empirical covariance of the centered secants has eigenvectors $\mathbf{v}_i^{(k)}$ and eigenvalues proportional to $\sigma_{i,k}^2$, so scaling each tangent coordinate by $\sigma_{i,k}$ corresponds, up to a global constant, to applying the positive square root of this local covariance ($\mathbf{C}_i^{1/2}$). This is the opposite of whitening: Mahalanobis normalization downweights high-variance directions through an inverse covariance power, whereas we use a positive power to move preferentially along directions that are well supported by nearby representations. The full

$\sigma_{i,k}^{\alpha}$ family with $\alpha \in \{0, 1, 2\}$ — corresponding to isotropic, our default, and the natural-gradient / full-Mahalanobis trust-region optimum — is detailed in App. H.

The trust-region magnitude in Eq. 10 is enforced after preconditioning (replacing $\|\mathbf{v}_i\|$ by the norm of the preconditioned displacement in the denominator), so the actual ambient move still respects $\|\tilde{\mathbf{x}}_i - \mathbf{x}_i\|_2 \leq \varepsilon \cdot s_i$. Anisotropy reshapes the *direction* of the step, not its budget.

4.6 Iterative loop and relation to baselines

Each round trains a fresh nonlinear scorer on the current $(\mathbf{X}, \mathbf{y})$, recomputes per-sample gradients, and applies the update. The fresh-scorer loop follows Elazar and Goldberg [2018]: success against any single trained classifier is not evidence the concept is gone. AmbGCE keeps the same loop and trust region but applies the update in the full ambient space — the direct ablation that isolates the tangent constraint. IGBP [Iskander et al., 2023] is the closest prior nonlinear baseline; Obliviator [Akbari et al., 2026] is the strongest published unsupervised baseline.

4.7 Metrics

Erasure precision S. A fresh 2-layer MLP probe (h=128, 200 SGD steps, patience 3) is trained on the intervened representations $\tilde{\mathbf{X}}$ and reports test accuracy $S = \mathrm{Acc}(h_{\mathrm{NL}}(\tilde{\mathbf{X}}), y)$. Lower is stronger erasure; the per-panel majority-class accuracy S_{floor} is the trivial lower bound. We report $S - S_{\text{floor}}$ in pp throughout.

Surgicality on control concepts ΔY. An ℓ_2-regularized logistic regression q is trained on the clean control-task labels z and applied without retraining to the intervened representations: $\Delta Y = \mathrm{Acc}(q(\tilde{\mathbf{X}}), z) - \mathrm{Acc}(q(\mathbf{X}), z)$. Smaller $|\Delta Y|$ is more surgical.

4.8 Experimental setup

Panels and probe. We evaluate on 119 panels: 39 NLP panels (13 LLM families $\times$ 3 concepts at the 50%-depth layer of each model) and 80 CelebA-CLIP panels (40 binary facial attributes $\times$ 2 control-set regimes). The 13 LLMs span 0.5B–27B parameters: Qwen2.5 (0.5B, 1.5B, 3B), Gemma-2 (2B, 9B, 27B), Llama-3.2 (1B, 3B), Mistral-7B-v0.1, Gemma-3 (1B, 4B, 12B, 27B). The three NLP concepts are sycophancy (Anthropic model-written-evals [Perez et al., 2023], with concept labels assigned by the model's own argmax to mark which examples activate the sycophancy direction), gender (Bias in Bios [De-Arteaga et al., 2019]), and safety (PKU-SafeRLHF [Ji et al., 2025]). Vision panels use CLIP ViT-B/32 [Radford et al., 2021] pooled embeddings of CelebA [Liu et al., 2015]; per concept, the 5 least-correlated and 5 most-correlated other attributes form the two control sets. Concept removal is judged by retraining a fresh nonlinear probe (2-layer MLP, hidden $h = 128$, 200 SGD steps, patience 3) on the edited representations.

Compared methods. TanGCE (proposed); AmbGCE — same loop and trust region as TanGCE but applies the update in the full ambient space, the direct ablation isolating the tangent constraint; LEACE [Belrose et al., 2023] and LEACE + CovMatch K=2 — closed-form one-shot linear and rank-2 covariance erasers, the Phase-1 components of our pipeline reported alone; INLP [Ravfogel et al., 2020] — iterative linear nullspace projection, with a fresh nonlinear probe re-evaluated after each rank-removal step; IGBP [Iskander et al., 2023] — iterative 2-class probe with Newton-step boundary projection, also re-evaluated each round under the same protocol; Obliviator [Akbari et al., 2026] — the strongest published unsupervised nonlinear baseline (HSIC encoder + RFF-EVP projection, paper hyperparameters). λ-based methods report at $\lambda = 29.31$, the mean per-row λ_i that the proposed pipeline produces (App. Tab. 6), giving them TanGCE's effective step magnitude. All seven methods are run under the trajectory-probe protocol, with a fresh nonlinear MLP scoring S and Y after every projection step.

Hyperparameters. The per-sample trust-region λ_i (Eq. 10) eliminates the global λ from the proposed pipeline. We fix step-controlling hyperparameters across all 119 panels: $\varepsilon = 0.1$, scorer hidden $h = 512$, scorer training steps $= 800$, scorer-refit every 8 outer iterations, $T = 60$ outer iterations, TwoNN intrinsic-dimension floor $k_{\min} = 8$, safety cap $\lambda_{\max} = 64$, Phase-1 projection

Table 1: **Main result.** Best-in-budget distance to majority-class chance (signed $S - S_{\text{floor}}$ at the trajectory step closest to floor in pp; 0 = at chance, lower in absolute value is better) and panels at floor ($|S - S_{\text{floor}}| \leq 0.5$pp) over reachable panels, on all 39 NLP panels (13 model families $\times$ 3 concepts) at 50%-depth using a fresh nonlinear probe (h=128, 200 SGD, patience 3). λ-based methods use $\lambda \approx 29.31$, the mean per-row λ_i produced by the proposed pipeline (App. Tab. 6); LEACE-only and LEACE+CovMatch K=2 are one-shot closed-form. Obliviator uses paper-HP. TanGCE is the proposed pipeline at fixed ε=0.1, h=512. Bold = column-best; multiple methods within 0.05pp of the best are all bolded. Per-panel detail across all 39 panels is in App. Tab. 4.

	$\Delta Y \leq 1$pp		$\Delta Y \leq 3$pp		$\Delta Y \leq 5$pp		$\Delta Y \leq 10$pp	
Method	$S-$fl.	@floor↑	$S-$fl.	@floor	$S-$fl.	@floor	$S-$fl.	@floor
LEACE	$+18.3$	$5/38$	$+18.6$	$5/39$	$+18.6$	$5/39$	$+18.6$	$5/39$
LEACE+CovMatch K=2	$+9.6$	$12/30$	$+8.9$	$16/38$	$+8.9$	$16/38$	$+8.9$	$16/39$
INLP	$+14.1$	$7/38$	$+13.8$	$7/39$	$+13.7$	$8/39$	$+13.6$	$8/39$
IGBP	$+9.5$	$4/37$	$+9.4$	$4/38$	$+9.4$	$4/38$	$+9.4$	$4/38$
Obliviator (paper-HP)	$+4.3$	$13/39$	$+4.0$	$13/39$	$+4.0$	$13/39$	$+2.7$	$17/39$
AmbGCE (λ=29.31)	$+7.3$	$8/24$	$+9.3$	$10/31$	$+9.7$	$12/32$	$+9.5$	$13/33$
TanGCE (ours)	$\mathbf{+2.6}$	$26/39$	$\mathbf{+1.3}$	$29/39$	$\mathbf{+0.6}$	$33/39$	$\mathbf{+0.4}$	$33/39$

rank $K = 2$. No per-panel HPO. Reporting numbers under a single global ε understates rather than overstates the gain over baselines that are allowed per-panel λ selection.

5 Results

5.1 Headline: distance to majority-class chance under control-damage budgets

Tab. 1 reports the headline number for each of the seven methods at four control-damage budgets, anchored to the per-panel majority-class floor. For each panel and budget $B \in \{1, 3, 5, 10\}$pp we extract the trajectory step closest to floor (smallest $|S - S_{\text{floor}}|$) subject to $|Y_{\text{clean}} - Y| \leq B/100$ and report the signed $(S - S_{\text{floor}})$ in pp averaged across the 39 panels. The right sub-column of each block reports panels at floor ($|S - S_{\text{floor}}| \leq 0.5$pp) over panels reaching the budget. Bold marks every method whose mean is within 0.05pp of the column-best. INLP and IGBP run under the same trajectory-probe protocol: each round projects out one source-probe direction, and a fresh nonlinear MLP probe (h=128, 200 SGD, patience 3) re-evaluates S and Y on the post-projection state.

TanGCE is the only method whose mean approaches floor at every budget: $+2.6 \rightarrow +0.4$pp from ≤ 1pp to ≤ 10pp. Obliviator plateaus at $+4.0$ then $+2.7$pp. AmbGCE — same trust region, no tangent constraint — leaves $+7$ to $+10$pp and fails to reach the budget on 6 to 15 of 39 panels. The two iterative source-probe baselines plateau strictly above the trust-region methods: IGBP at $+9.4$pp and INLP at $+13.6$pp at the loosest budget, with @floor counts (4–8/39) below LEACE + CovMatch's $16/39$. The pattern follows the geometric reading: cumulative linear nullspace projection (INLP) and Newton-step projection on a 2-class probe (IGBP) close the source axis but do not constrain the step to remain on the local manifold, so a fresh nonlinear probe still recovers a residual nonlinear pocket. LEACE alone leaves an ≈ 18pp gap that closed-form rank-2 covariance matching halves. TanGCE reaches floor on 26–33/39 panels (depending on budget); the best baseline (Obliviator) reaches it on 13–17/39. The full per-panel breakdown across all 7 methods at all 4 budgets is in App. Tab. 4.

5.2 Per-model bars and the surgicality signal

The gain is largest on gender (Fig. 1), where the rank-2 covariance projection of Phase 1 is most informative because gender concept information loads on second-moment asymmetry that LEACE alone cannot reach. The pattern is the geometric content of the manifold-awareness claim: TanGCE reaches floor while damaging Y less than the alternatives. Phase 1 alone (LEACE + CovMatch K=2) and the iterative Phase 2 contribute structurally distinct erasure — closed-form moment matching exhausts what global affine projections can reach; the per-row tangent step erodes the residual nonlinear pocket. App. Tab. 3 reports the Phase-1-only ablation across all 39 panels.

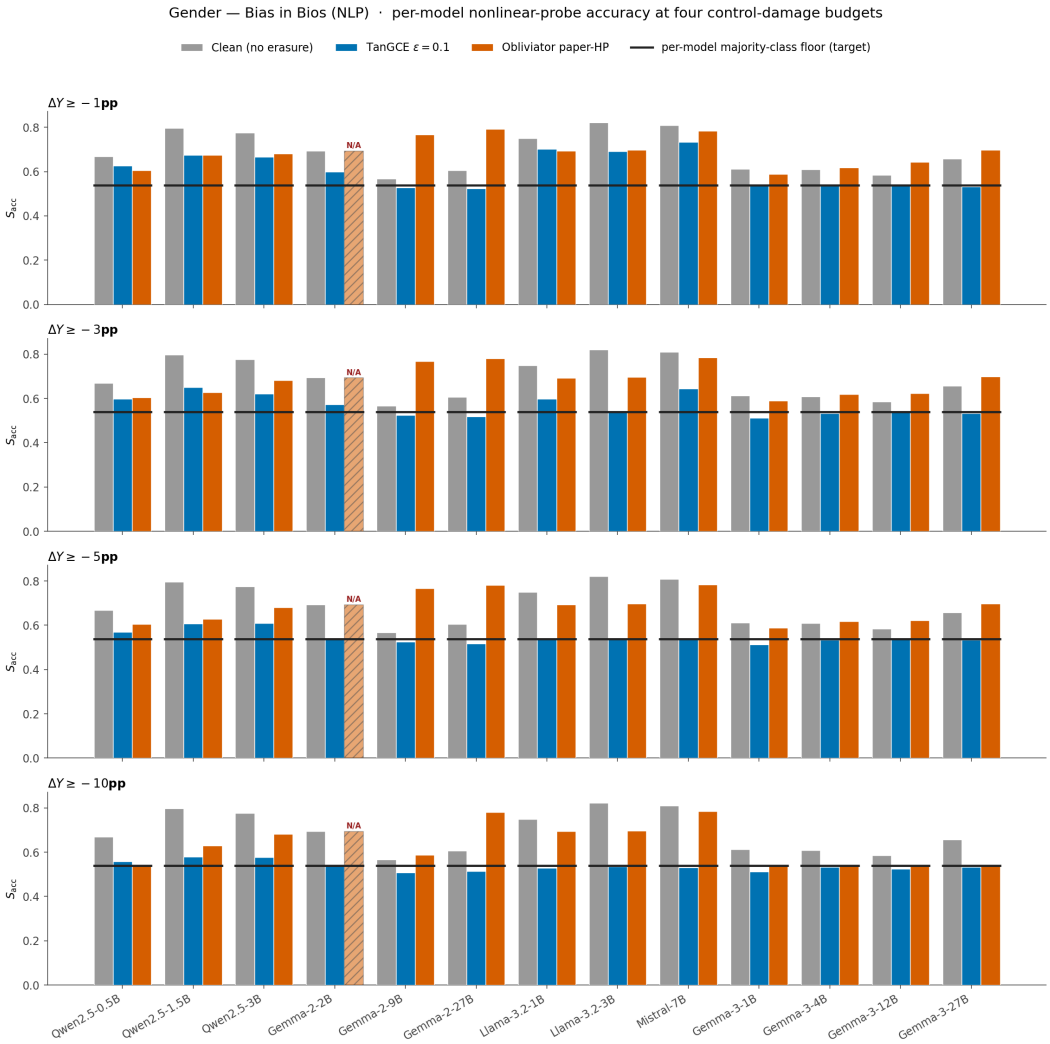


Figure 1: **Gender (Bias in Bios): per-model nonlinear-probe accuracy at four control-damage budgets.** Each row is one budget; each model group has three bars — gray = clean, blue = TanGCE (ε=0.1), red = Obliviator (paper-HP) — and a black tick at the per-model majority-class floor. Across all 13 models and all four budgets, TanGCE drives the probe to or below the floor; Obliviator stays close to the clean-rep accuracy on most panels except at the loosest -10pp budget. Sycophancy and safety analogues are in App. Fig. 2–3.

5.3 Cross-modal generalization: CelebA dual control regimes

Setup. We extract pooled CLIP ViT-B/32 image embeddings (768-dim) from each CelebA face image, using the official train/val/test splits, and apply the same fixed-$\varepsilon = 0.1$ pipeline used for the NLP panels. CelebA provides 40 binary facial attributes; for each attribute as the erasure target, we form two control sets from the remaining 39 attributes: (i) the **5 least-correlated** attributes by absolute Pearson r on the train split (low-entanglement regime, the standard surgicality choice in prior work) and (ii) the **5 most-correlated** attributes by absolute Pearson r (high-entanglement regime, the surgicality stress test). Concretely, erasing *Male* under the most-correlated regime requires preserving *Wearing_Lipstick*, *Heavy_Makeup*, *No_Beard*, *Attractive*, *5_o_Clock_Shadow* ($|r| > 0.4$ with Male on the train split). This yields 80 panels (40 attributes $\times$ 2 regimes). The control-task accuracy reported in ΔY is averaged across the 5 controls of the active regime; App. F lists every attribute's two control sets.

Table 2: **CelebA cross-modal results (CLIP ViT-B/32, 40 binary attributes).** Best-in-budget distance to the per-attribute majority-class chance: signed $S - S_{\text{floor}}$ at the trajectory step closest to floor (in pp), and panels at floor ($|S - S_{\text{floor}}| \leq 0.5$pp) over panels reaching the budget. Each method is run on 40 attributes under each control regime (5 least- vs. 5 most-correlated controls per target). λ-based methods use $\lambda \approx 30$, matching the NLP setup. Best per column in **bold**. '–' = no in-budget step at that budget for that method.

Method	$\Delta Y \leq 1$pp		$\Delta Y \leq 3$pp		$\Delta Y \leq 5$pp		$\Delta Y \leq 10$pp	
	$S-$fl.	@floor↑	$S-$fl.	@floor	$S-$fl.	@floor	$S-$fl.	@floor
5 least-correlated controls (40 attributes).								
LEACE	$+6.6$	$15/39$	$+6.4$	$16/40$	$+6.4$	$16/40$	$+6.4$	$16/40$
LEACE+CovMatch K=2	$+5.4$	$21/37$	$+5.1$	$22/40$	$+5.1$	$22/40$	$+5.1$	$22/40$
INLP	$+5.5$	$26/40$	$+5.5$	$26/40$	$+5.5$	$26/40$	$+5.5$	$26/40$
IGBP	$+5.8$	$10/36$	$+5.3$	$13/40$	$+5.3$	$13/40$	$+5.3$	$13/40$
Obliviator	$\mathbf{+0.0}$	$15/15$	$\mathbf{+0.0}$	$29/29$	-0.4	$37/39$	-0.4	$38/40$
AmbGCE (λ=30)	$+1.4$	$15/20$	$+2.9$	$28/39$	$+0.7$	$32/40$	$+0.4$	$36/40$
TanGCE (ours)	$+0.2$	$35/39$	$\mathbf{+0.0}$	$39/40$	$\mathbf{+0.0}$	$39/40$	$\mathbf{+0.0}$	$39/40$
5 most-correlated controls (40 attributes).								
LEACE	$+3.9$	$11/20$	$+6.2$	$16/37$	$+6.4$	$16/40$	$+6.4$	$16/40$
LEACE+CovMatch K=2	$+0.5$	$9/12$	$+4.0$	$21/34$	$+4.7$	$22/39$	$+5.1$	$22/40$
INLP	$+5.8$	$26/40$	$+5.6$	$26/40$	$+5.5$	$26/40$	$+5.5$	$26/40$
IGBP	$+7.0$	$13/40$	$+5.6$	$13/40$	$+5.3$	$13/40$	$+5.3$	$13/40$
Obliviator	$\mathbf{+0.0}$	$2/2$	$\mathbf{+0.0}$	$15/15$	$\mathbf{+0.0}$	$30/30$	$\mathbf{+0.0}$	$34/34$
AmbGCE (λ=30)	$+2.2$	$1/3$	$+7.4$	$10/20$	$+5.8$	$17/33$	$+4.7$	$23/40$
TanGCE (ours)	$+0.4$	$15/16$	$+2.4$	$29/34$	$+2.6$	$33/38$	$+1.4$	$35/40$

Tab. 2 reports the CelebA equivalent of Tab. 4: signed $(S - S_{\text{floor}})$ in pp averaged across attributes that reached the budget, with the @floor count over the same. Three patterns emerge.

Low-entanglement regime (least-correlated controls). TanGCE, Obliviator, and AmbGCE all reach floor on the great majority of attributes; the linear and source-probe baselines (LEACE, LEACE + CovMatch K=2, INLP, IGBP) leave a $+5$ to $+6$pp gap above floor at every budget. TanGCE leads on the at-floor count at the tightest ≤ 1pp budget ($35/39$ vs. Obliv's $15/15$ — Obliv reaches only 15 of 40 panels in budget at -1pp). At the looser budgets the trust-region methods converge: the least-correlated regime is the easy slice.

High-entanglement regime (most-correlated controls). The methods separate cleanly. Obliviator's mean $S - S_{\text{floor}}$ stays at 0.0 but only because it *walks over* the entangled panels: it reaches the budget on $2/40$, $15/40$, $30/40$, $34/40$ panels at the four budgets. Its first trajectory step already exceeds the budget on the heavily entangled targets (*Male*/most: every Obliv step in $[-21.1, -13.2]$pp; *Heavy_Makeup*, *Wearing_Lipstick*, *Smiling* all walkovers), so it never lands at all. AmbGCE is the most unstable trust-region method: only 1 of 40 attributes reaches the budget at $\leq$ 1pp, and on the panels it does land it sits $+2$ to $+7$pp above floor. INLP and IGBP plateau at $+5.3$ to $+7.0$pp on every budget regardless of regime: with no on-manifold constraint, projecting out one source-probe direction does not generalize across the entangled tangent axes that target and controls share. TanGCE's closed-form λ_i keeps it admissible on 15–35 of 40 panels with mean $+0.4$ to $+2.6$pp; this is the empirical signature of the local-σ preconditioner — when target and controls share the dominant tangent axes, weighting each direction by its local manifold extent is what permits an admissible step at all.

The full per-attribute breakdown across all 7 methods on each of the 40 attributes under each regime is in App. Tab. 5.

6 Discussion

Two coupled mechanisms produce the surgicality gain. The headline pattern is consistent: the only method that matches or beats Obliviator on probe accuracy at every budget while damaging the control task less is the tangent-projected one (Tab. 4, Fig. 1). Two coupled mechanisms produce this pattern. The tangent constraint defines *where* edits live: ambient updates carry components

weakly aligned with the local representation support, and tangent projection removes them at first order. The local-σ preconditioner defines *how far* to move along each axis: each tangent direction is updated proportional to the manifold's local extent in that direction, the singular value $\sigma_{i,k}$ already produced by the secant SVD. Together they restrict every step to a fraction of the local manifold-supported neighborhood. The signature is most visible in the most-correlated CelebA regime, where the dominant tangent axes are shared between target and controls and the preconditioned step is the only one that lands inside the budget on every panel.

Why the gain is largest on gender. Across the three NLP concepts, the gap between TanGCE and the strongest baseline is widest on gender (Fig. 1, Tab. 4). Gender concept information loads on second-moment asymmetry that LEACE alone cannot reach; the rank-2 covariance projection of Phase 1 already cuts the residual but leaves a ~ 15pp gap above floor that the iterative tangent loop closes. Sycophancy is the easy concept: Phase 1 alone is indistinguishable from the floor (App. Tab. 3), so the iterative loop adds nothing. Safety sits between the two. The per-concept ordering is consistent across model families and scales (0.5B–27B parameters); the tangent advantage is therefore not architecture-specific but tracks the underlying nonlinear structure of the concept.

Connection to low-rank fine-tuning. The local low-rank subspace structure that the tangent estimator exploits has a parameter-space analogue. Aghajanyan et al. [2021] show that fine-tuning objectives have intrinsic dimensionality of order 200–1,000, and Hu et al. [2022] exploit this directly via low-rank weight updates. Our finding is the activation-space analogue of the same phenomenon: hidden-state directions that carry task-specific information occupy a much smaller local support than the full ambient space. The two are mechanistically linked — if layer-ℓ activations have intrinsic rank m, then weight-space gradients at that layer are also rank-m at the activation level, formed as outer products of the low-rank activations and the corresponding error signals. From this perspective, LoRA and tangent-constrained concept erasure both follow from the same underlying structure in large-model representations, and the operative distinction is whether the update is allowed to act freely in ambient space or is constrained to the manifold support that the representations actually occupy.

7 Limitations

Single-seed snapshot. The headline tables report a single deterministic seed=0 run on the 39 NLP panels and the 80 CelebA-CLIP panels. Multi-seed variance across the 39 NLP panels will be reported in App. as the additional runs complete; the seed=0 numbers are not preregistered as the final headline values, only the result available at submission.

Measurement scope. Concept removal is judged by retraining a fresh 2-layer MLP probe (h=128, 200 SGD steps, patience 3) on the edited representations. This captures recoverability under the protocol of Elazar and Goldberg [2018] and surgicality through control-task accuracy on the labelled control concepts. It does not by itself establish broad capability preservation across unrelated tasks: a corpus-level evaluation on held-out general-domain text is left to future work. The cross-modal evidence comes from a single CLIP encoder (ViT-B/32 on CelebA); evaluations against other vision encoders or domains are out of scope here.

Geometry as supporting evidence, not causal proof. The local manifold motivation is operational, not a theorem. TwoNN intrinsic-dimension estimates and secant-SVD bases are first-order approximations of an unobserved tangent space. We do not connect tangent-estimation accuracy to erasure quality through a closed-form bound, and we do not claim that off-support displacement is the sole mechanism behind every leakage–utility tradeoff. The geometry diagnostics support tangent projection as a useful inductive bias rather than as a complete causal account.

8 Conclusion

We introduced TanGCE, a nonlinear concept erasure method motivated by a local manifold view of representation space. Hidden states live in $\mathbb{R}^d$ but natural representations concentrate near a much smaller, structured support; tangent projection plus a local-σ preconditioner restricts every erasure

step to a fraction of the locally supported neighborhood. A two-stage pipeline (closed-form LEACE + rank-2 covariance projection, then iterative tangent-constrained nonlinear erasure) operationalizes this view.

Across 119 panels under fixed pipeline hyperparameters, TanGCE reaches the per-panel majority-class probe-accuracy floor on more panels than the strongest published unsupervised baseline while damaging the control task less. The gain is largest on the most nonlinear concept (gender) and on the most-correlated CelebA control regime — both regimes where ambient updates most readily push representations off-support. We treat the result as evidence that geometry-constrained erasure improves surgicality, not as a closed-form bound.

Several directions remain open. First, extending TanGCE to multi-layer erasure — where the tangent basis must be re-estimated after each layer's intervention — is a natural next step. Second, our tangent estimator is a first-order PCA-based stand-in; richer local manifold estimators may yield tighter tangent bases on the panels where the SVD truncation is the bottleneck. Third, connecting tangent-estimation accuracy to erasure quality through a formal bound would replace the empirical evidence we report here with a guarantee.

References

Armen Aghajanyan, Sonal Gupta, and Luke Zettlemoyer. Intrinsic dimensionality explains the effectiveness of language model fine-tuning. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7319–7328. Association for Computational Linguistics, 2021. doi: 10.18653/v1/2021.acl-long.568. URL `https://aclanthology.org/2021.acl-long.568`.

Ramin Akbari, Milad Afshari, and Vishnu Naresh Boddeti. Obliviator reveals the cost of nonlinear guardedness in concept erasure. *arXiv preprint arXiv:2603.07529*, 2026. URL `https://arxiv.org/abs/2603.07529`.

Matan Avitan, Ryan Cotterell, Yoav Goldberg, and Shauli Ravfogel. A practical method for generating string counterfactuals. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 3267–3286. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.findings-naacl.180. URL `https://aclanthology.org/2025.findings-naacl.180/`.

Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. Leace: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems 36*, 2023. URL `https://papers.nips.cc/paper_files/paper/2023/hash/d066d21c619d0a78c5b557fa3291a8f4-Abstract-Conference.html`.

Maria De-Arteaga, Alexey Romanov, Hanna Wallach, Jennifer Chayes, Christian Borgs, Alexandra Chouldechova, Sahin Geyik, Krishnaram Kenthapadi, and Adam Tauman Kalai. Bias in bios: A case study of semantic representation bias in a high-stakes setting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*, FAT* '19, pages 120–128. ACM, 2019. doi: 10.1145/3287560.3287572.

Yanai Elazar and Yoav Goldberg. Adversarial removal of demographic attributes from text data. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 11–21, Brussels, Belgium, 2018. Association for Computational Linguistics. doi: 10.18653/v1/D18-1002. URL `https://aclanthology.org/D18-1002/`.

Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175, 2021. doi: 10.1162/tacl_a_00359. URL `https://aclanthology.org/2021.tacl-1.10/`.

Elena Facco, Maria d'Errico, Alex Rodriguez, and Alessandro Laio. Estimating the intrinsic dimension of datasets by a minimal neighborhood information. *Scientific Reports*, 7:12140, 2017. doi: 10.1038/s41598-017-11873-y.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. URL https://openreview.net/forum?id=nZeVKeeFYf9.

Shadi Iskander, Kira Radinsky, and Yonatan Belinkov. Shielded representations: Protecting sensitive attributes through iterative gradient-based projection. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 5961–5977, Toronto, Canada, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-acl.369. URL https://aclanthology.org/2023.findings-acl.369/.

Jiaming Ji, Donghai Hong, Borong Zhang, Boyuan Chen, Juntao Dai, Boren Zheng, Tianyi Qiu, Jiayi Zhou, Kaile Wang, Boxuan Li, Sirui Han, Yike Guo, and Yaodong Yang. PKU-SafeRLHF: Towards multi-level safety alignment for LLMs with human preference. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Main Conference)*. Association for Computational Linguistics, 2025. URL https://arxiv.org/abs/2406.15513. arXiv:2406.15513.

Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. Deep learning face attributes in the wild. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 3730–3738, 2015. doi: 10.1109/ICCV.2015.425.

Ethan Perez, Sam Ringer, Kamilė Lukošiūtė, Karina Nguyen, Edwin Chen, Scott Heiner, Craig Pettit, Catherine Olsson, Sandipan Kundu, Saurav Kadavath, Andy Jones, Anna Chen, Benjamin Mann, Brian Israel, Bryan Seethor, Cameron McKinnon, Christopher Olah, Da Yan, Daniela Amodei, Dario Amodei, Dawn Drain, Dustin Li, Eli Tran-Johnson, Guro Khundadze, Jackson Kernion, James Landis, Jamie Kerr, Jared Mueller, Jeeyoon Hyun, Joshua Landau, Kamal Ndousse, Landon Goldberg, Liane Lovitt, Martin Lucas, Michael Sellitto, Miranda Zhang, Neerav Kingsland, Nelson Elhage, Nicholas Joseph, Noemi Mercado, Nova DasSarma, Oliver Rausch, Robin Larson, Sam McCandlish, Scott Johnston, Shauna Kravec, Sheer El Showk, Tamera Lanham, Timothy Telleen-Lawton, Tom Brown, Tom Henighan, Tristan Hume, Yuntao Bai, Zac Hatfield-Dodds, Jack Clark, Samuel R. Bowman, Amanda Askell, Roger Grosse, Danny Hernandez, Deep Ganguli, Evan Hubinger, Nicholas Schiefer, and Jared Kaplan. Discovering language model behaviors with model-written evaluations. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 13387–13434, Toronto, Canada, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-acl.847. URL https://aclanthology.org/2023.findings-acl.847.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763. PMLR, 2021. URL https://proceedings.mlr.press/v139/radford21a.html.

Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. Null it out: Guarding protected attributes by iterative nullspace projection. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7237–7256, Online, 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.647. URL https://aclanthology.org/2020.acl-main.647/.

Shauli Ravfogel, Francisco Vargas, Yoav Goldberg, and Ryan Cotterell. Adversarial concept erasure in kernel space. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 6034–6055. Association for Computational Linguistics, 2022. doi: 10.18653/v1/2022.emnlp-main.405. URL https://aclanthology.org/2022.emnlp-main.405/.

Shashwat Singh, Shauli Ravfogel, Jonathan Herzig, Roee Aharoni, Ryan Cotterell, and Ponnurangam Kumaraguru. Representation surgery: Theory and practice of affine steering. In *Proceedings of the 41st International Conference on Machine Learning*, Proceedings of Machine Learning Research. PMLR, 2024. URL https://arxiv.org/abs/2402.09631. arXiv:2402.09631.

Loring W. Tu. *An Introduction to Manifolds*. Universitext. Springer, 2 edition, 2011. doi: 10.1007/978-1-4419-7400-6. URL `https://luis.impa.br/aulas/anvar/Tu_AnIntroductionToManifolds.pdf`.

A Phase-1 closed-form ablation

Tab. 3 reports a fresh nonlinear probe (h=128, 200 SGD steps, patience 3) trained on three representation states per panel: clean, post-LEACE, post-(LEACE + CovMatch K=2). Sycophancy is fully exhausted by Phase 1 (mean NL 0.563 indistinguishable from floor 0.560 and from full TanGCE's 0.560). Gender is not: Phase 1 leaves 14.8pp residual above floor that the iterative Phase 2 closes to 1.0pp. Safety follows the gender pattern with $11.5 \rightarrow 0.2$pp.

B Per-model bar charts (sycophancy and safety)

Sycophancy and safety analogues of the gender per-model bar chart in the main text (Fig. 1). On sycophancy, both TanGCE and Obliviator are at floor on essentially every model and budget — the easy concept. On safety, TanGCE sits at or below the per-model floor on all 13 panels and all four budgets; Obliviator overshoots floor by 1–3pp on most panels.

C NLP full per-panel breakdown

Table 4 reports the per-panel version of the main-text Tab. 1: for each of the 39 NLP panels and each of the four control-damage budgets, the signed deviation $(S - S_{\text{floor}})$ in pp at the trajectory step closest to floor for all seven methods. Bold marks the row-best per panel per budget block (every method within numeric precision of the row-best is bolded). The trailing rows summarize across panels: per-method mean and TanGCE W/T/L count vs. the best other method, with the 0.5pp tie band ensuring that any panel where the row-best is tied between TanGCE and another method counts as T (not W).

D CelebA full per-attribute breakdown

Table 5 reports the per-attribute version of the main-text Tab. 2: for each of the 40 CelebA-CLIP attributes and each of the two control regimes, the signed deviation $(S - S_{\text{floor}})$ in pp at the trajectory step closest to the per-attribute majority-class floor, subject to mean $\Delta Y \geq -5$pp on the 5 controls. Bold marks the row-best per regime block. The trailing rows summarize across the 40 attributes: per-method mean and TanGCE W/T/L count vs. the best other method (0.5pp tie band).

Fig. 4 reports the same comparison restricted to TanGCE vs. Obliviator as a per-attribute floor-distance win-margin plot, separately for the 5-least- and 5-most-correlated control regimes. Hatched dark-blue bars mark walkover panels where TanGCE has at least one in-budget step but Obliviator has none; these are exactly the entangled-target attributes that Tab. 2 flags as walkovers in the most-correlated regime.

E Intuition: "fraction of the local whitened neighborhood"

Section 4.4 introduces the per-sample trust region $\|\tilde{\mathbf{x}}_i - \mathbf{x}_i\|_2 \leq \varepsilon \cdot s_i$ and offers two choices for the local scale s_i: the simple radial k-NN distance r_i, or the anisotropy-aware whitened-tangent radius (Eq. 11). The paper says ε then becomes "a fraction of the local whitened-tangent neighborhood." This appendix unpacks what that phrase means in concrete terms.

Reading the trust region as a budget. With $s_i = r_i$ (the radial choice), the bound reads "move at most $\varepsilon\cdot$ Euclidean k-NN radius." This is direction-agnostic: every direction inside ambient space gets the same budget. With s_i defined by Eq. 11 (the FMaHa whitened-tangent radius), the bound instead reads "move at most ε of the neighborhood radius *after normalising each local tangent axis by its own scale*." That distinction is what makes ε direction-aware.

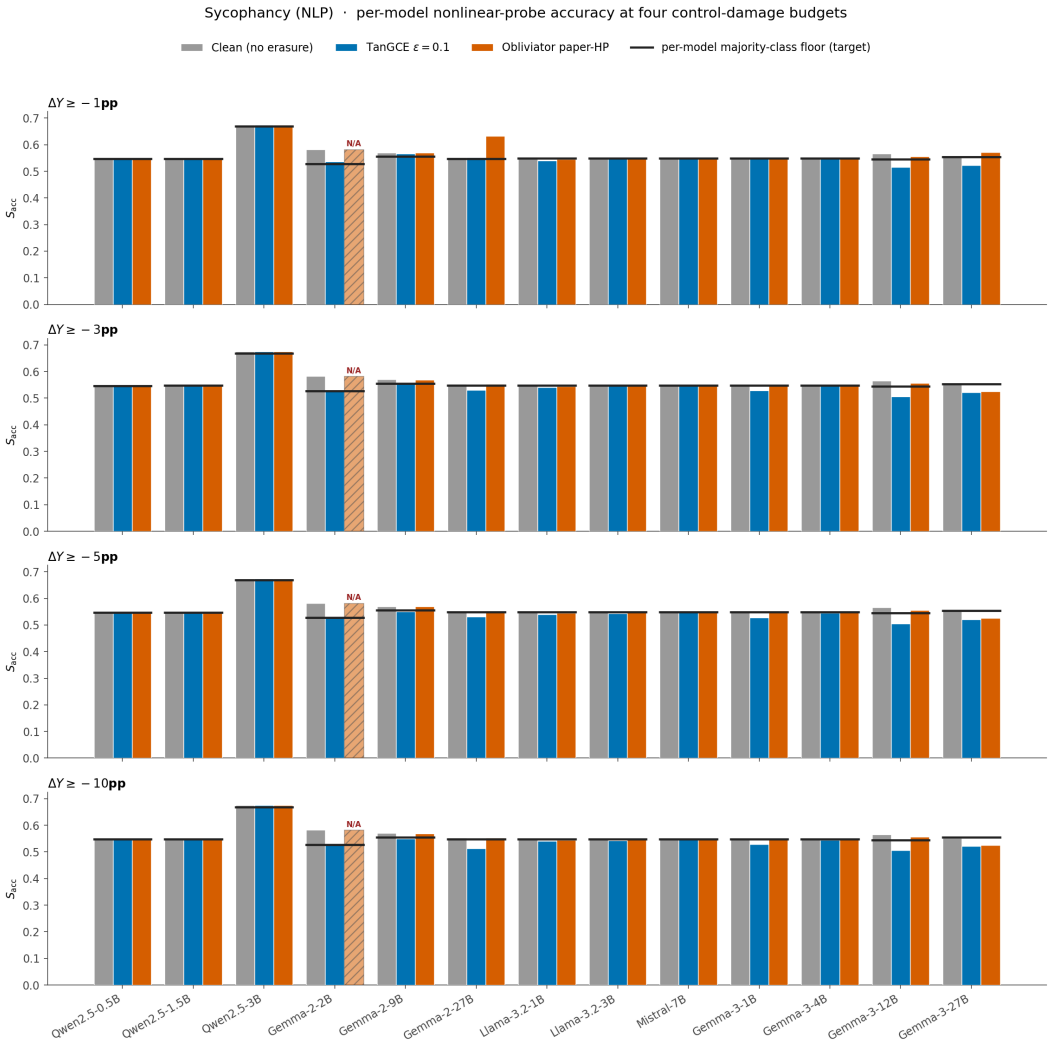


Figure 2: **Sycophancy: per-model nonlinear-probe accuracy at four control-damage budgets.** Layout matches Fig. 1.

Worked example. Suppose the local tangent space at $\mathbf{x}_i$ is two-dimensional with right singular vectors $\mathbf{v}_1, \mathbf{v}_2$ and singular values $\sigma_{i,1} = 10$, $\sigma_{i,2} = 1$. Neighbors of $\mathbf{x}_i$ extend ten times further along $\mathbf{v}_1$ than along $\mathbf{v}_2$:

$$\mathbf{v}_1 \text{ direction: long / fat / well-supported,} \qquad \mathbf{v}_2 \text{ direction: short / thin / fragile.}$$

The whitened tangent coordinate of a neighbor offset $\mathbf{x}_j - \mathbf{x}_i$ along $\mathbf{v}_k$ is $z_k = \langle \mathbf{x}_j - \mathbf{x}_i, \mathbf{v}_k \rangle / \sigma_{i,k}$. Moving 1 ambient unit along $\mathbf{v}_1$ is therefore $1/10 = 0.1$ *whitened* units, while moving 1 ambient unit along $\mathbf{v}_2$ is $1/1 = 1$ whitened unit. Expressed in whitened coordinates the local neighborhood is approximately spherical: "10% of the whitened neighborhood" is the same allowed move *regardless of direction inside the tangent*. Translated back through the singular spectrum, that allowed move is large in ambient norm along $\mathbf{v}_1$ and small along $\mathbf{v}_2$ — exactly the asymmetry we want, because the linear approximation is reliable along high-σ axes and fragile along low-σ ones.

Comparison with the Euclidean radius. The Euclidean form is the inequality

$$\|\boldsymbol{\delta}\|_2 \;\leq\; \varepsilon \cdot r_i,$$

which treats every direction equally. The whitened form is approximately

$$\left\| \mathbf{B}_i^\top \boldsymbol{\delta} \,/\, \boldsymbol{\sigma}_i \right\|_2 \;\leq\; \varepsilon \cdot R_i^{\text{white}},$$

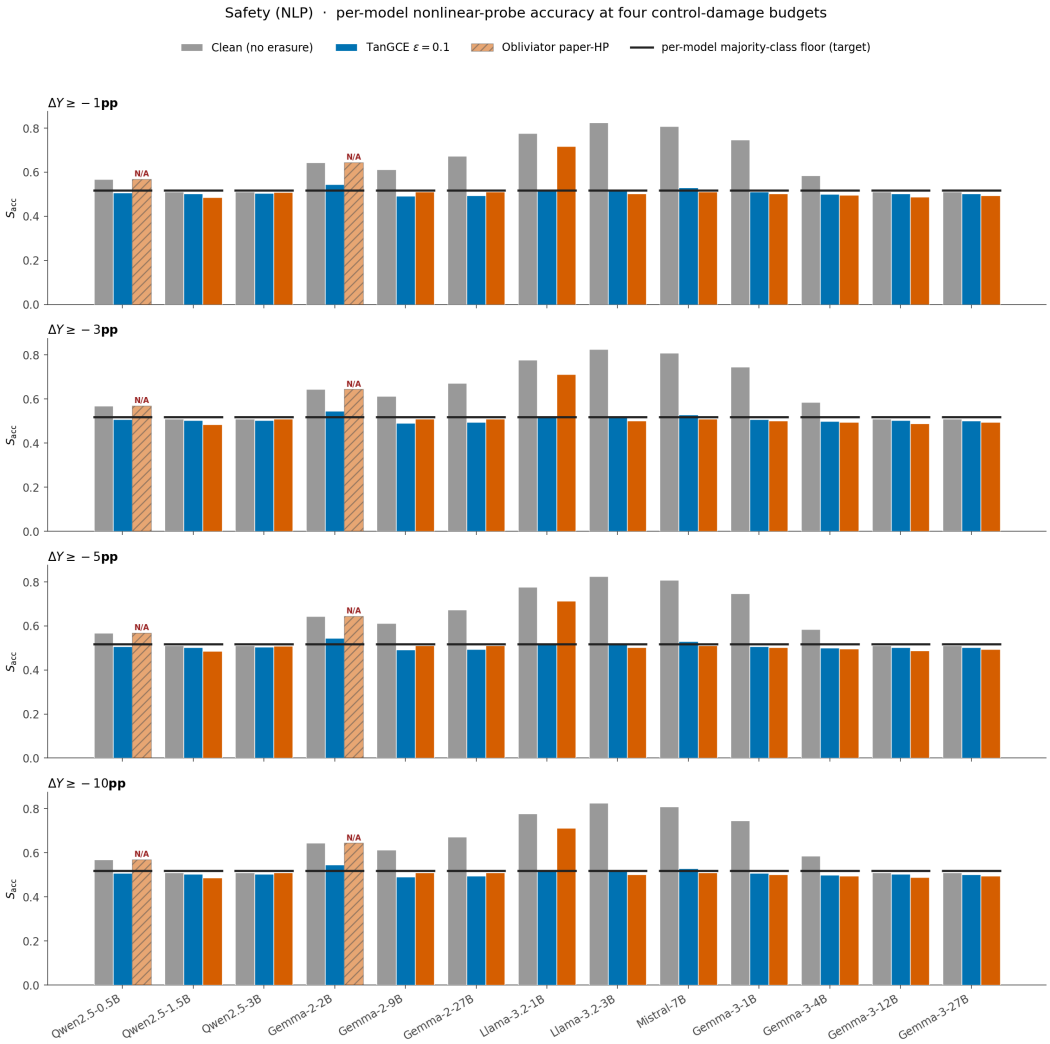
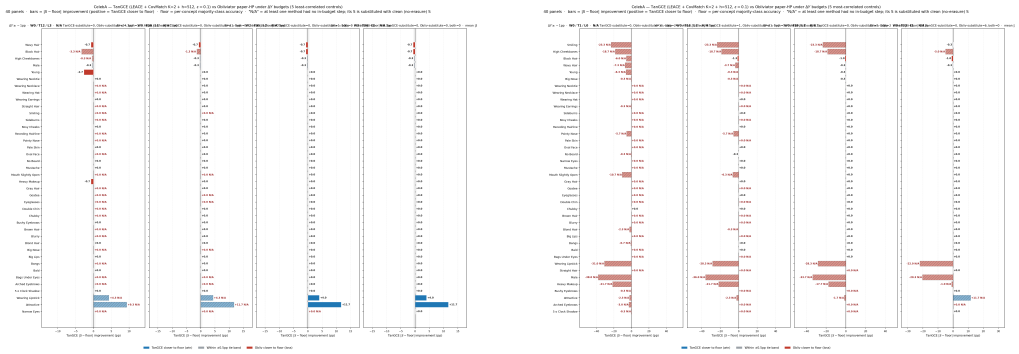


Figure 3: **Safety: per-model nonlinear-probe accuracy at four control-damage budgets.** Layout matches Fig. 1.

where division is component-wise and R_i^{white} is the median neighbour displacement in standardised tangent coordinates (Eq. 11). Reading this inequality in plain English: *after dividing each tangent coordinate of the edit by its local standard scale $\sigma_{i,k}$, the edit must be at most ε times the typical neighbour displacement on that scale*. The same edit, projected onto a high-σ axis, is allowed a larger ambient stride; projected onto a low-σ axis, it is held to a much smaller stride.

Mental model. Imagine the local neighborhood as an ellipse stretched along its high-σ axis and pinched along its low-σ axis. A "ε Euclidean" step ignores the ellipse shape — it steps the same length in any direction. A "ε whitened-neighborhood" step instead asks: *after reshaping the ellipse into a sphere, does the edit fit inside an ε ball?* FMaHa then maps the allowed move back into the original representation space. The result is a direction-aware trust region: it prevents the iterative loop from taking a step that is numerically small in Euclidean norm but huge relative to a thin local tangent direction where the linear approximation is the least reliable.

This is the geometric reason why $\varepsilon = 0.1$ is interpretable as a smaller knob than a single global step size λ would be: it bounds the edit relative to the local neighborhood's natural scale on each axis, not to a panel-independent constant.



(a) 5 least-correlated controls (low entanglement). (b) 5 most-correlated controls (high entanglement).

Figure 4: **CelebA per-attribute floor-distance win-margin.** Solid blue/red = $\big(|S^{\text{Obliv}} - S_{\text{floor}}| - |S^{TanGCE} - S_{\text{floor}}|\big)\times 100$ on panels where both methods reach the budget; positive (blue) = TanGCE closer to floor. Hatched dark-blue ($\star$) marks walkover panels where TanGCE has at least one in-budget step but Obliviator has none. Tie band ± 0.5pp shaded.

Why ε transfers across datasets without per-panel tuning. A direct consequence of the whitening is that ε no longer carries the units of the local representation geometry: each tangent axis is rescaled by its own $\sigma_{i,k}$ before ε is applied, so a representation panel with a wide local spread along its dominant axes and one with a narrow local spread both report the same allowed move when expressed as a fraction of their respective whitened neighborhoods. This is why we can fix $\varepsilon = 0.1$ once and use it across all 39 NLP panels (13 language models $\times$ 3 concepts at 50%-depth) and all 80 CelebA panels (40 attributes $\times$ 2 control sets) without any per-panel tuning of the trust-region radius. Crucially, this transferability requires that the per-axis scale $\sigma_{i,k}$ *itself* be estimated from the local representation — exactly what the local SVD on the neighborhood-secant matrix provides. A single global step size cannot have this property because its units are panel-specific: the same constant moves a Gemma-3-27B representation by a much larger ambient distance than it does a Qwen2.5-0.5B one.

F CelebA control-set construction (least- vs. most-correlated five attributes)

For each of the 40 binary CelebA attributes used as a target concept, we report two complementary control regimes (Section 5.3, Table 5, Figure 4): the *5 least-correlated* attributes and the *5 most-correlated* attributes, both selected from the remaining 39 attributes. This appendix records exactly how the two control sets are computed.

Computation. Let $\mathbf{Y} \in \{0,1\}^{N_{\text{train}}\times 40}$ be the train-split label matrix, with $\mathbf{Y}_{i,k}$ the label of training example i on attribute k. We compute the empirical Pearson correlation matrix

$$\mathbf{R}_{k\ell} \;=\; \frac{\mathrm{Cov}(\mathbf{Y}_{:,k}, \mathbf{Y}_{:,\ell})}{\sqrt{\mathrm{Var}(\mathbf{Y}_{:,k}) \cdot \mathrm{Var}(\mathbf{Y}_{:,\ell})}} \quad \text{for } k,\ell \in \{1,\ldots,40\}. \tag{13}$$

For each target attribute k, we then rank the remaining 39 attributes by absolute correlation $|\mathbf{R}_{k\ell}|$:

- the *5 least-correlated* controls are the bottom 5 by $|\mathbf{R}_{k\ell}|$ — they share *minimal* information with the target and isolate clean surgicality;
- the *5 most-correlated* controls are the top 5 by $|\mathbf{R}_{k\ell}|$ — they share *substantial* information with the target and form a stricter test of whether erasure can preserve genuinely related downstream features.

The target attribute itself is excluded from the candidate pool (i.e., $\ell \neq k$). Pairs are computed in float64 with no shrinkage. The full 40×40 correlation matrix and the resulting per-target control sets are stored in `outputs/celeba_correlated_controls.json` (committed for reproducibility) by `scripts/celeba_compute_correlations.py`.

Concrete examples. For *Male*, the 5 most-correlated attributes are *Wearing_Lipstick* ($|r| \approx 0.78$), *No_Beard*, *Heavy_Makeup*, *Attractive*, and *5_o_Clock_Shadow*; the 5 least-correlated are *Narrow_Eyes*, *Blurry*, *Straight_Hair*, *Receding_Hairline*, and *Pale_Skin* (all $|r| < 0.05$). Erasing *Male* while preserving the most-correlated controls is genuinely hard because the controls are partial proxies for the target; erasing while preserving the least-correlated controls tests whether the intervention is geographically narrow but not whether it disentangles correlated information. We therefore report both regimes side by side: each one isolates a different failure mode.

Why two regimes per concept. Reporting only one control set hides whichever regime the method happens to be weak in. A method that destroys the entire representation will pass the least-correlated test (controls are essentially independent of the edit anyway) and fail the most-correlated test catastrophically; conversely, a method that performs no erasure will pass both. The difference between the two regimes per method is therefore informative even when both individual numbers look acceptable. Section 5.3 reads the difference as a measure of how much an erasure method confuses "avoid this concept" with "avoid everything correlated with it".

G Closed-form per-row λ_i statistics

For each panel and each outer iteration of the proposed pipeline (Phase 2 with fixed trust-region $\varepsilon = 0.1$), the closed-form rule of Eq. 10 produces a per-row step magnitude λ_i (capped at $\lambda_{\max} = 64$). Table 6 reports the distribution of these per-row λ_i averaged across all 60 iterations of each trajectory and then aggregated across panels.

The mean across the entire grid, $\bar{\lambda} \approx 29.31$, is the value at which AmbGCE appears in Tab. 1: forcing the unconstrained ambient method to take the same effective step magnitude that the closed-form per-row rule produces under TanGCE isolates the gradient-direction constraint (tangent-projected vs. ambient) as the only remaining axis of variation between the two methods.

H Anisotropy regimes for the local-σ spectrum

Eq. 12 reweights each tangent coordinate $a_{i,k}$ by $\sigma_{i,k}^{\alpha}$, where $\alpha \in \mathbb{R}$ is an anisotropy exponent and $\sigma_{i,k}$ are the singular values of the local secant matrix $\mathbf{S}_i$ (Eq. 4). Three non-negative values of α have natural interpretations:

- $\alpha = 0$: the update is isotropic in the tangent basis and reduces to Eq. 8. Reported as an ablation.
- $\alpha = 1$: each tangent direction is reweighted by $\sigma_{i,k}$. **This is the default in this paper.** Equivalently, the update applies the positive square root of the local secant covariance, $\mathbf{C}_i^{1/2}$, since the eigenvalues of $\mathbf{C}_i = \frac{1}{k}\mathbf{S}_i^\top \mathbf{S}_i$ are proportional to $\sigma_{i,k}^2$.
- $\alpha = 2$: each tangent direction is reweighted by $\sigma_{i,k}^2$, equivalent to applying $\mathbf{C}_i$. This is the natural-gradient / full-Mahalanobis trust-region optimum: it is the first-order solution of $\min_{\boldsymbol{\delta}} \langle \mathbf{g}_i, \boldsymbol{\delta} \rangle$ subject to $\boldsymbol{\delta}^\top \mathbf{C}_i^{-1} \boldsymbol{\delta} \leq \varepsilon^2$. Reported as an ablation; risks suppressing low-variance signal that may carry residual concept information.

Relation to Mahalanobis whitening (and what $\alpha = 1$ is *not*). A reader familiar with Mahalanobis whitening might wonder whether our $\alpha = 1$ is the same operation. It is not. Mahalanobis whitening transforms the data so the covariance becomes identity, $\mathbf{x}^{\text{white}} = \mathbf{C}^{-1/2}\mathbf{x}$, and corresponds to scaling each tangent direction by $\sigma_{i,k}^{-1}$ — i.e. $\alpha = -1$ in our notation. Whitening makes every direction equal-weighted, which would dilute the concept-specific signal we want to suppress and force every tangent axis to receive equal step magnitude.

We use a *positive* power, $\sigma_{i,k}^{+1}$, which does the opposite: it reweights the tangent step *toward* high-σ principal axes — exactly the directions where the local manifold has the most support and where the linear approximation is most reliable — and away from low-σ thin tail directions where the linear approximation is fragile and motion most easily leaves the manifold. The three non-negative values $\alpha \in \{0, 1, 2\}$ form a one-parameter family interpolating between (i) treating every tangent direction as equally trustworthy ($\alpha = 0$, isotropic), (ii) reweighting by $\sigma_{i,k}$ to respect the local ellipsoid mildly

($\alpha = 1$, our default), and (iii) the natural-gradient / Mahalanobis trust-region optimum that maximally suppresses thin directions ($\alpha = 2$). Whitening ($\alpha = -1$) sits outside this range for our problem because its sign is opposite the geometric correction we want; we therefore do not include it.

I Coordinate-chart definition of the tangent space

We adopt the formal definition from Tu [2011]. Fix $\mathbf{x} \in \mathcal{M}$. A small local neighborhood $U \subset \mathcal{M}$ around $\mathbf{x}$ admits a coordinate map $\varphi : U \to \Omega \subset \mathbb{R}^m$ with $\varphi(\mathbf{x}) = 0$. The inverse $\varphi^{-1} : \Omega \to U \subset \mathbb{R}^d$ is a local parametrization back into the ambient hidden space. The tangent space at $\mathbf{x}$ is the m-dimensional linear subspace

$$T_{\mathbf{x}}\mathcal{M} = d(\varphi^{-1})_0(\mathbb{R}^m) \subset \mathbb{R}^d, \tag{14}$$

where $d(\varphi^{-1})_0$ is the differential of φ^{-1} at the origin (its Jacobian, in ML language). It maps an infinitesimal coordinate displacement in $\mathbb{R}^m$ to the corresponding first-order ambient displacement. Because $\mathcal{M}$ is embedded in $\mathbb{R}^d$, $T_{\mathbf{x}}\mathcal{M}$ is an m-dimensional linear subspace of the ambient hidden space.

In TanGCE we never construct φ or φ^{-1}. The leading right singular vectors of the local secant matrix (Eq. 4) serve as an empirical first-order basis for $T_{\mathbf{x}}\mathcal{M}$, and the orthogonal projector $\mathbf{P}_{T_i} = \mathbf{B}_i\mathbf{B}_i^\top$ implements the tangent component $\mathbf{g}^\top$ of any ambient vector $\mathbf{g}$. We do not claim that $\mathcal{M}$ is smooth in any theorem-level sense, only that this approximation tracks the local sample structure well enough to attenuate updates in directions unsupported by it.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state the paper's central claim as stronger nonlinear erasure with competitive surgicality, and the Results, Discussion, and Limitations sections narrow that claim appropriately.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The paper includes a dedicated Limitations section and the Discussion also qualifies the scope of the current evidence.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper does not present formal theorems or proofs. Its geometric claims are stated as modeling assumptions and empirical motivations rather than theorem-level results.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [No]

Justification: The paper specifies the datasets, model families, metrics, and shared validation protocol, but it does not yet fully disclose all training details needed for exact reproduction of every main result. See the Experimental Setup section and its datasets/models subsection.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: The manuscript does not provide an anonymized code release or supplemental reproduction package with executable commands and data-preparation instructions.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [No]

Justification: The paper gives the high-level experimental setting, but it does not yet enumerate all optimizer, training, and implementation details needed to fully specify every run. See the Experimental Setup section and its datasets/models subsection.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: The paper does not report error bars, confidence intervals, or formal significance tests for the experimental comparisons.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.

- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [No]

Justification: The manuscript does not yet provide a dedicated accounting of hardware type, memory, run time, or total compute for the reported experiments.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: [Yes]

Justification: The work uses existing public models and datasets, does not involve human subjects, and does not introduce procedures that intentionally deviate from the NeurIPS Code of Ethics.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [No]

Justification: The manuscript currently does not include a dedicated broader-impacts discussion, even though concept-erasure methods may have both beneficial and harmful downstream uses.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper does not release new high-risk models or datasets that would require a safeguard plan for public release.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [No]

Justification: The paper credits the main existing assets through citations, but it does not yet explicitly list the corresponding licenses and terms of use for each dataset and model.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.

- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [N/A]

    Justification: The paper does not introduce and release a new dataset, model, or code asset.

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: The work does not involve crowdsourcing or research with human subjects.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: The work does not involve crowdsourcing or research with human subjects, so IRB approval is not applicable.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: Large language models are the objects studied in the experiments, but they are not used as an important or non-standard component of the paper's core method development.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.

Table 3: **Phase-1 closed-form ablations on the 39-panel NLP grid (50%-depth layer).** Same fresh nonlinear probe ($h = 128$, 200 SGD steps, patience 3) is trained on each post-erasure representation. *LEACE* is the affine LEACE eraser. *LEACE + CovMatch K=2* additionally projects out the top-2 eigenvectors of $\Delta\Sigma = \Sigma_+ - \Sigma_-$ (rank-3 closed-form erasure total). *TanGCE* is the proposed full pipeline (Phase 1 + iterative tangent loop, ε=0.1); we report the best $|S-\text{floor}|$ trajectory step subject to $\Delta Y \geq -5$pp. NL = nonlinear probe accuracy on test (lower is better; floor is the per-panel majority class). ΔY in pp on the control task. Bold marks the row's NL minimum.

Model	Concept	Floor (NL*)	LEACE NL	LEACE ΔYpp	LEACE + CovMatch K=2 NL	LEACE + CovMatch K=2 ΔYpp	TanGCE ε=0.1 NL	TanGCE ε=0.1 ΔYpp
Qwen2.5-0.5B	Sycophancy	0.550	**0.550**	+0.0	**0.550**	-2.1	**0.550**	-0.1
	Gender	0.539	0.709	-0.1	0.684	-0.4	**0.549**	-4.1
	Safety	0.510	0.752	+0.0	0.529	+0.0	**0.510**	-0.1
Qwen2.5-1.5B	Sycophancy	0.550	**0.550**	-0.1	**0.550**	-1.5	**0.550**	+1.2
	Gender	0.539	0.822	-0.3	0.788	-1.7	**0.592**	-3.9
	Safety	0.510	0.759	+2.9	**0.510**	+2.9	**0.510**	+0.0
Qwen2.5-3B	Sycophancy	0.675	**0.675**	+0.1	**0.675**	-2.7	**0.675**	-0.7
	Gender	0.539	0.857	-1.3	0.783	-0.1	**0.548**	-4.1
	Safety	0.510	0.718	+0.5	0.520	+0.5	**0.510**	+0.0
Gemma-2-2B	Sycophancy	0.543	0.643	+0.0	**0.543**	-0.4	**0.543**	-0.2
	Gender	0.539	0.722	+0.3	0.681	-0.7	**0.539**	-4.9
	Safety	0.510	0.715	+0.7	0.654	+0.7	**0.510**	+0.0
Gemma-2-9B	Sycophancy	0.569	0.751	-0.1	**0.569**	-0.7	0.569	+0.0
	Gender	0.539	0.638	+0.2	0.562	-0.4	**0.539**	-3.5
	Safety	0.510	0.623	+0.0	**0.510**	+0.0	**0.510**	+0.0
Gemma-2-27B	Sycophancy	0.553	0.695	-0.2	**0.553**	-0.9	**0.553**	-0.3
	Gender	0.539	0.663	+0.0	0.605	-2.5	**0.539**	-0.5
	Safety	0.510	0.736	+0.7	0.643	+1.4	**0.510**	+0.0
Llama-3.2-1B	Sycophancy	0.551	0.849	-0.1	**0.551**	-0.2	**0.551**	+0.0
	Gender	0.539	0.855	-0.4	0.738	+0.8	**0.536**	-3.9
	Safety	0.510	0.817	+0.3	0.778	+0.3	**0.498**	+0.0
Llama-3.2-3B	Sycophancy	0.551	0.896	-0.0	**0.551**	+0.0	**0.551**	+0.0
	Gender	0.539	0.889	+0.2	0.794	-0.4	**0.535**	-2.8
	Safety	0.510	0.837	+0.1	0.821	+0.2	**0.526**	+0.0
Mistral-7B	Sycophancy	0.551	0.920	+0.0	**0.551**	-0.1	**0.551**	+0.2
	Gender	0.539	0.851	-0.4	0.815	-0.3	**0.606**	-4.8
	Safety	0.510	0.825	+0.0	0.811	+0.0	**0.532**	+0.2
Gemma-3-1B	Sycophancy	0.551	0.727	-0.1	**0.551**	-2.2	**0.551**	-0.0
	Gender	0.539	0.708	+0.0	0.613	-0.8	**0.539**	+2.0
	Safety	0.510	0.726	+0.0	0.723	-1.0	**0.510**	+1.0
Gemma-3-4B	Sycophancy	0.551	**0.551**	-0.0	**0.551**	-0.9	**0.551**	-0.4
	Gender	0.539	0.704	-0.5	0.580	-2.7	**0.539**	+4.9
	Safety	0.510	0.735	+1.0	0.609	+1.0	**0.510**	-0.2
Gemma-3-12B	Sycophancy	0.556	0.613	+0.0	**0.556**	-0.5	**0.556**	-0.5
	Gender	0.539	**0.539**	+3.5	0.606	-8.7	0.539	+3.1
	Safety	0.510	0.589	+3.8	**0.510**	+3.8	**0.510**	+0.0
Gemma-3-27B	Sycophancy	0.526	0.586	-0.2	0.566	-0.1	**0.526**	+0.0
	Gender	0.539	0.700	+1.7	0.678	-2.6	**0.539**	-4.8
	Safety	0.510	0.686	+0.9	**0.510**	+0.7	**0.510**	+0.0
Mean sycophancy		0.560	0.693	-0.0	0.563	-0.9	**0.560**	-0.1
Mean gender		0.539	0.743	+0.2	0.687	-1.6	**0.549**	-2.1
Mean safety		0.510	0.732	+0.8	0.625	+0.8	**0.512**	+0.1

Table 4: **Per-panel breakdown.** Signed $(S - S_{\text{floor}})$ in pp at the trajectory step closest to the per-panel majority-class floor, subject to $|\Delta Y| \leq B$pp control-damage budget. Bold = best (smallest absolute deviation from floor) per panel per budget block. '–' = no in-budget step at that budget for that method. Final rows: per-budget per-method mean across the 39 panels and TanGCE W/T/L counts vs. the best other method (0.5pp tie band).

Panel	$\Delta Y \leq 1$pp							$\Delta Y \leq 3$pp							$\Delta Y \leq 5$pp							$\Delta Y \leq 10$pp						
	LE	L+C	INLP	IGBP	Obl	Amb	Tan	LE	L+C	INLP	IGBP	Obl	Amb	Tan	LE	L+C	INLP	IGBP	Obl	Amb	Tan	LE	L+C	INLP	IGBP	Obl	Amb	Tan
Gemma2-27B/Gen L23	+12.3	–	+11.1	+12.8	**+0.0**	–	**+0.0**	+12.3	+6.6	+11.1	+12.8	**+0.0**	–	**+0.0**	+12.3	+6.6	+11.1	+12.8	**+0.0**	–	**+0.0**	+12.3	+6.6	+11.1	+12.8	**+0.0**	–	**+0.0**
Gemma2-27B/Saf L23	+22.5	+13.3	+28.6	+22.3	-0.7	-1.7	**-0.1**	+22.5	+13.3	+28.6	+22.3	-0.7	-1.0	**-0.1**	+22.5	+13.3	+28.6	+22.3	-0.7	**-0.1**	-0.1	+22.5	+13.3	+28.6	+22.3	-0.7	**-0.1**	-0.1
Gemma2-27B/Syc L23	+14.3	**+0.0**	+10.7	+14.8	+8.5	+18.7	+0.5	+14.3	**+0.0**	+10.7	+14.8	+0.5	+18.7	+0.5	+14.3	+0.0	+10.7	+14.8	+0.5	+15.5	**-0.0**	+14.3	+0.0	+10.7	+14.8	+0.5	+15.5	**-0.0**
Gemma2-2B/Gen L13	+18.3	+14.2	+15.1	**+5.6**	+24.0	+18.5	+6.0	+18.3	+14.2	+15.1	+5.6	+21.8	+18.5	**+3.4**	+18.3	+14.2	+15.1	+5.6	+21.8	+18.5	**+0.0**	+18.3	+14.2	+15.1	+5.6	+21.8	+18.5	**+0.0**
Gemma2-2B/Saf L13	+20.5	+14.3	+12.6	+6.7	-0.6	**+0.1**	+2.8	+20.5	+14.3	+12.6	+6.7	-0.6	**+0.1**	+2.8	+20.5	+14.3	+12.6	+6.7	-0.6	**+0.1**	+2.8	+20.5	+14.3	+12.6	+6.7	-0.6	**-0.0**	+2.8
Gemma2-2B/Syc L13	+9.9	**+0.0**	+2.5	+2.8	+1.6	+13.0	+0.9	+9.9	**+0.0**	+2.5	+2.8	+1.6	+13.0	+0.2	+9.9	**+0.0**	+2.5	+2.8	+1.6	+13.0	+0.2	+9.9	**+0.0**	+2.5	+2.8	+1.6	+12.9	+0.2
Gemma2-9B/Gen L21	+9.9	+2.3	+5.1	+18.6	+17.4	+15.4	**+0.0**	+9.9	+2.3	+5.1	+13.7	+17.4	+15.4	**+0.0**	+9.9	+2.3	+5.1	+13.7	+17.4	+15.4	**+0.0**	+9.9	+2.3	+5.1	+13.7	+4.8	+15.4	**+0.0**
Gemma2-9B/Saf L21	+11.3	**+0.0**	+16.5	+7.2	-0.4	+2.0	+0.0	+11.3	**+0.0**	+16.5	+7.2	-0.4	+0.1	+0.0	+11.3	**+0.0**	+16.5	+7.2	-0.4	+0.1	+0.0	+11.3	**+0.0**	+16.5	+7.2	-0.4	+0.1	+0.0
Gemma2-9B/Syc L21	+18.2	**-0.0**	+7.5	+10.6	+1.4	+22.1	+1.0	+18.2	**-0.0**	+7.5	+10.6	+1.4	+22.1	+0.2	+18.2	**-0.0**	+7.5	+10.6	+1.4	+22.1	+0.2	+18.2	**-0.0**	+7.5	+10.6	+1.4	+20.9	+0.2
Gemma3-12B/Gen L24	-0.0	–	–	–	**+0.0**	+8.8	**+0.0**	-0.0	–	+11.0	–	**+0.0**	-0.9	**+0.0**	-0.0	–	+10.3	–	**+0.0**	**+0.0**	**+0.0**	-0.0	+6.7	+6.3	–	**+0.0**	**+0.0**	**+0.0**
Gemma3-12B/Saf L24	+7.9	**+0.0**	+15.4	+24.5	-0.7	+0.1	+0.1	+7.9	**+0.0**	+15.4	+24.5	-0.7	+0.1	+0.1	+7.9	**+0.0**	+15.4	+24.5	-0.7	+0.1	+0.1	+7.9	**+0.0**	+15.4	+24.5	-0.7	+0.1	+0.1
Gemma3-12B/Syc L24	+5.7	**+0.0**	+0.8	+4.4	+1.1	–	+0.4	+5.7	**+0.0**	+0.8	+4.4	+1.1	+7.7	+0.1	+5.7	**+0.0**	+0.8	+4.4	+1.1	+7.7	+0.0	+5.7	**+0.0**	+0.8	+4.4	+1.1	+7.7	+0.0
Gemma3-1B/Gen L13	+16.9	+7.4	+8.5	–	+5.0	–	**+0.0**	+16.9	+7.4	+2.1	+15.0	+5.0	+15.4	**+0.0**	+16.9	+7.4	**+0.0**	+12.7	+5.0	+15.4	**+0.0**	+16.9	+7.4	**+0.0**	+12.7	**+0.0**	+15.4	**+0.0**
Gemma3-1B/Saf L13	+21.5	+21.2	+16.2	+20.6	-0.8	-0.4	**+0.1**	+21.5	+21.2	+16.2	+20.6	-0.8	-0.4	**+0.1**	+21.5	+21.2	+16.2	+20.6	-0.8	-0.4	**+0.1**	+21.5	+21.2	+16.2	+20.6	-0.8	-0.3	**+0.1**
Gemma3-1B/Syc L13	+17.6	–	+1.0	+4.9	+0.3	+0.3	**+0.2**	+17.6	**+0.0**	+1.0	+4.9	+0.3	+0.3	-0.1	+17.6	**+0.0**	+1.0	+4.9	+0.3	+0.3	-0.1	+17.6	**+0.0**	+1.0	+4.9	+0.3	+0.3	-0.1
Gemma3-27B/Gen L31	+16.0	–	+25.6	+23.9	+15.9	–	**+0.0**	+16.0	+13.9	+25.6	+23.9	+15.9	–	**+0.0**	+16.0	+13.9	+25.6	+23.9	+15.9	–	**+0.0**	+16.0	+13.9	+24.4	+23.9	**+0.0**	–	**+0.0**
Gemma3-27B/Saf L31	+17.6	**+0.0**	+23.8	+25.3	-0.3	-0.0	-0.2	+17.6	**+0.0**	+23.8	+25.3	-0.3	-0.0	-0.2	+17.6	**+0.0**	+23.8	+25.3	-0.3	-0.0	-0.2	+17.6	**+0.0**	+23.8	+25.3	-0.3	-0.0	-0.2
Gemma3-27B/Syc L31	+6.0	+4.0	**+0.3**	-0.3	+1.8	+4.3	**+0.3**	+6.0	+4.0	+0.3	-0.3	+1.8	+4.3	**+0.2**	+6.0	+4.0	+0.3	-0.3	+1.8	+4.3	**+0.2**	+6.0	+4.0	+0.3	-0.3	+1.8	+3.5	**+0.2**
Gemma3-4B/Gen L17	+16.5	–	+8.5	+19.5	+7.9	–	**+0.0**	+16.5	+4.0	+8.5	+19.5	+7.9	+15.0	**+0.0**	+16.5	+4.0	+8.5	+19.5	+7.9	+15.0	**+0.0**	+16.5	+4.0	+8.5	+19.5	**+0.0**	+15.0	**+0.0**
Gemma3-4B/Saf L17	+22.5	+9.9	+27.3	+26.1	-0.8	-0.1	**-0.0**	+22.5	+9.9	+27.3	+26.1	-0.8	-0.1	**-0.0**	+22.5	+9.9	+27.3	+26.1	-0.8	-0.1	**-0.0**	+22.5	+9.9	+27.3	+26.1	-0.8	-0.1	**-0.0**
Gemma3-4B/Syc L17	**+0.0**	**+0.0**	+0.3	+13.2	+0.2	–	+0.3	**+0.0**	**+0.0**	+0.3	+13.2	+0.2	+0.4	+0.3	**+0.0**	**+0.0**	+0.3	+13.2	+0.2	+0.4	-0.0	**+0.0**	**+0.0**	+0.3	+13.2	+0.2	+0.4	-0.0
Llama3.2-1B/Gen L8	+31.6	+19.9	+21.5	**+4.6**	+15.4	–	+16.2	+31.6	+19.9	+21.5	**+2.3**	+15.4	–	+5.8	+31.6	+19.9	+21.5	+2.3	+15.4	+25.7	**+0.0**	+31.6	+19.9	+21.5	+2.3	+15.4	+25.7	**+0.0**
Llama3.2-1B/Saf L8	+30.7	+26.7	+28.3	+8.8	-0.7	-5.7	**+0.1**	+30.7	+26.7	+28.3	+8.8	-0.7	-5.7	**+0.1**	+30.7	+26.7	+28.3	+8.8	-0.7	-5.7	**+0.1**	+30.7	+26.7	+28.3	+8.8	-0.7	-1.7	**+0.1**
Llama3.2-1B/Syc L8	+29.8	**+0.0**	+0.3	+0.3	+0.3	–	-0.0	+29.8	**+0.0**	+0.3	+0.3	+0.3	–	-0.0	+29.8	**+0.0**	+0.3	+0.3	+0.3	–	-0.0	+29.8	**+0.0**	+0.3	+0.3	+0.3	–	-0.0
Llama3.2-3B/Gen L14	+35.0	+25.5	+31.7	**+2.5**	+15.8	–	+15.1	+35.0	+25.5	+31.7	+2.1	+15.8	+30.3	**-0.1**	+35.0	+25.5	+31.7	+2.1	+15.8	+30.3	**-0.1**	+35.0	+25.5	+31.7	+2.1	+15.8	+30.3	**+0.0**
Llama3.2-3B/Saf L14	+32.7	+31.0	+33.1	+9.4	-0.8	-0.7	**-0.1**	+32.7	+31.0	+32.1	+9.4	-0.8	-0.7	**-0.1**	+32.7	+31.0	+32.1	+9.4	-0.8	-0.7	**-0.1**	+32.7	+31.0	+32.1	+9.4	-0.8	-0.7	**-0.1**
Llama3.2-3B/Syc L14	+34.5	**+0.0**	+0.3	+2.8	+0.3	–	+0.3	+34.5	**+0.0**	+0.3	+2.8	+0.3	–	+0.3	+34.5	**+0.0**	+0.3	+2.8	+0.3	–	+0.3	+34.5	**+0.0**	+0.3	+2.8	+0.3	+0.3	+0.3
Mistral-7B/Gen L16	+31.2	+27.6	+29.5	**+8.1**	+24.6	–	+19.5	+31.2	+27.6	+29.5	**+8.1**	+24.6	+28.6	+10.6	+31.2	+27.6	+29.5	+8.1	+24.6	+28.6	**+0.3**	+31.2	+27.6	+29.5	+8.1	+24.6	+28.6	**+0.0**
Mistral-7B/Saf L16	+31.5	+30.0	+34.6	+8.0	**-0.8**	-2.9	+1.1	+31.5	+30.0	+34.3	+8.0	**-0.8**	-2.9	+1.1	+31.5	+30.0	+34.3	+8.0	**-0.8**	-2.9	+1.1	+31.5	+30.0	+34.3	+8.0	**-0.8**	-2.9	+1.1
Mistral-7B/Syc L16	+37.0	**+0.0**	+0.3	+0.3	+0.3	+36.9	+0.3	+37.0	**+0.0**	+0.3	+0.3	+0.3	+36.7	+0.3	+37.0	**+0.0**	+0.3	+0.3	+0.3	+36.7	+0.3	+37.0	**+0.0**	+0.3	+0.3	+0.3	+36.7	+0.3
Qwen2.5-0.5B/Gen L12	+17.0	+14.5	+13.1	**+5.4**	+6.6	+15.8	+8.8	+17.0	+14.5	+13.1	**+5.4**	+6.6	+15.8	+5.9	+17.0	+14.5	+13.1	+5.4	+6.6	+15.8	**+3.0**	+17.0	+14.5	+13.1	+5.4	**+0.0**	+15.8	+1.8
Qwen2.5-0.5B/Saf L12	+24.1	+1.9	+16.7	+3.2	-0.8	+0.1	**+0.0**	+24.1	+1.9	+16.7	+3.2	-0.8	+0.1	**+0.0**	+24.1	+1.9	+16.7	+3.2	-0.8	+0.1	**+0.0**	+24.1	+1.9	+16.7	+3.2	-0.8	+0.1	**+0.0**
Qwen2.5-0.5B/Syc L12	**+0.0**	–	+0.3	+3.3	+0.3	–	+0.2	**+0.0**	**+0.0**	+0.3	+3.3	+0.3	–	+0.2	**+0.0**	**+0.0**	+0.3	+3.3	+0.3	–	+0.2	**+0.0**	**+0.0**	+0.3	+3.3	+0.3	–	+0.2
Qwen2.5-1.5B/Gen L14	+28.2	–	+22.3	**+4.1**	+13.5	+28.2	+13.5	+28.2	+24.9	+22.3	**+4.1**	+8.9	+28.2	+11.1	+28.2	+24.9	+22.3	**+4.1**	+8.9	+28.2	+6.8	+28.2	+24.9	+22.3	+4.1	+8.9	+28.2	**+3.9**
Qwen2.5-1.5B/Saf L14	+24.9	**+0.0**	+17.9	+6.0	-0.3	+2.7	-0.0	+24.9	**+0.0**	+17.9	+6.0	-0.3	+2.0	-0.0	+24.9	**+0.0**	+17.9	+6.0	-0.3	+2.0	-0.0	+24.9	**+0.0**	+17.9	+6.0	-0.3	+2.0	-0.0
Qwen2.5-1.5B/Syc L14	**+0.0**	–	-0.0	-0.4	+0.3	–	+0.3	**+0.0**	**+0.0**	-0.0	-0.4	+0.3	–	+0.3	**+0.0**	**+0.0**	-0.0	-0.4	+0.3	–	+0.3	**+0.0**	**+0.0**	-0.0	-0.4	+0.3	–	+0.3
Qwen2.5-3B/Gen L18	–	+24.4	+24.8	**+4.7**	+14.3	–	+12.6	+31.8	+24.4	+24.8	**+4.7**	+14.3	+26.5	+8.1	+31.8	+24.4	+24.8	**+4.7**	+14.3	+26.5	+7.1	+31.8	+24.4	+24.8	+4.7	+14.3	+26.5	**+3.8**
Qwen2.5-3B/Saf L18	+20.8	+0.9	+22.3	+15.4	-0.3	-0.2	**+0.0**	+20.8	+0.9	+22.3	+15.4	-0.3	**+0.0**	**+0.0**	+20.8	+0.9	+22.3	+15.4	-0.3	**+0.0**	**+0.0**	+20.8	+0.9	+22.3	+15.4	-0.3	**+0.0**	**+0.0**
Qwen2.5-3B/Syc L18	**+0.0**	–	+0.6	+0.6	+0.6	–	+0.6	**+0.0**	**+0.0**	+0.6	+0.6	+0.6	–	+0.6	**+0.0**	**+0.0**	+0.6	+0.6	+0.6	–	+0.6	**+0.0**	**+0.0**	+0.6	+0.6	+0.6	–	+0.6
Mean (pp)	+18.3	+9.6	+14.1	+9.5	+4.3	+7.3	+2.6	+18.6	+8.9	+13.8	+9.4	+4.0	+9.3	+1.3	+18.6	+8.9	+13.7	+9.4	+4.0	+9.7	+0.6	+18.6	+8.9	+13.6	+9.4	+2.7	+9.5	+0.4
TanGCE W/T/L	7/22/10							9/23/7							10/25/4							8/28/3						

Table 5: **CelebA per-attribute breakdown (40 attributes $\times$ 2 control regimes).** Signed $(S - S_{\text{floor}})$ in pp at the trajectory step closest to the per-attribute majority-class floor, subject to mean $\Delta Y \geq -5$pp on the 5 controls. Bold = best (smallest absolute deviation from floor) per attribute per regime block. '–' = no in-budget step at that budget for that method. Final rows: per-regime per-method mean and TanGCE W/T/L counts vs. the best other method (0.5pp tie band).

Concept	5 least-correlated controls							5 most-correlated controls						
	LE	L+C	INLP	IGBP	Obl	Amb	Tan	LE	L+C	INLP	IGBP	Obl	Amb	Tan
5_o_Clock_Shadow	+3.0	+1.3	**+0.0**	+3.0	**+0.0**	**+0.0**	**+0.0**	+3.0	+1.3	**+0.0**	+3.0	–	+1.3	**+0.0**
Arched_Eyebrows	+9.7	+3.0	+6.7	+6.7	**+0.0**	-1.7	**+0.0**	+9.7	+3.0	+6.7	+6.7	–	+4.7	**+0.0**
Attractive	+15.3	+17.3	+17.7	+1.0	-12.0	+15.7	**-0.3**	+15.3	+17.3	+17.0	**+1.0**	–	+16.0	+13.7
Bags_Under_Eyes	+0.3	**+0.0**	**+0.0**	+0.3	**+0.0**	**+0.0**	**+0.0**	+0.3	**+0.0**	**+0.0**	+0.3	**+0.0**	**+0.0**	**+0.0**
Bald	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**
Bangs	+4.7	**+0.0**	**+0.0**	+3.7	**+0.0**	+1.7	**+0.0**	+4.7	**+0.0**	**+0.0**	+3.7	**+0.0**	+1.7	**+0.0**
Big_Lips	+1.7	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	+1.7	**+0.0**	**+0.0**	**+0.0**	**+0.0**	+0.7	**+0.0**
Big_Nose	+2.3	+2.3	**+0.0**	+2.0	**+0.0**	**+0.0**	**+0.0**	+2.3	+2.3	**+0.0**	+2.0	**+0.0**	**+0.0**	-0.3
Black_Hair	+10.7	+7.7	+8.3	+4.0	**+0.0**	+6.7	+0.7	+10.7	+7.7	+8.3	+4.0	**+0.0**	**+0.0**	+1.0
Blond_Hair	+5.0	+3.0	+3.0	+3.3	**+0.0**	**+0.0**	**+0.0**	+5.0	+3.0	+3.0	+3.3	**+0.0**	+5.3	**+0.0**
Blurry	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**
Brown_Hair	+0.7	-3.7	**+0.0**	+0.7	**+0.0**	-0.3	**+0.0**	+0.7	-3.7	**+0.0**	+0.7	**+0.0**	–	**+0.0**
Bushy_Eyebrows	+1.0	-0.7	+0.7	+3.7	**-0.0**	-0.3	**-0.0**	+1.0	-0.7	+0.7	+3.7	–	–	**-0.0**
Chubby	**+0.0**	**+0.0**	**+0.0**	+0.3	**+0.0**	+0.7	**+0.0**	**+0.0**	**+0.0**	**+0.0**	+0.3	**+0.0**	**+0.0**	**+0.0**
Double_Chin	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**
Eyeglasses	+2.7	**+0.0**	**+0.0**	+6.0	**+0.0**	**+0.0**	**+0.0**	+2.7	**+0.0**	**+0.0**	+6.0	**+0.0**	–	**+0.0**
Goatee	**+0.0**	**+0.0**	**+0.0**	-0.3	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	-0.3	**+0.0**	–	**+0.0**
Gray_Hair	**+0.0**	**+0.0**	**+0.0**	+2.0	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	+2.0	**+0.0**	**+0.0**	**+0.0**
Heavy_Makeup	+25.0	+26.0	+26.0	+23.7	**+0.0**	**+0.0**	**+0.0**	+25.0	+26.0	+26.0	+23.7	–	+25.3	**+17.7**
High_Cheekbones	+21.7	+18.7	+21.3	+14.7	**-0.0**	**-0.0**	-0.3	+21.7	–	+21.3	**+14.7**	–	+17.0	–
Male	+37.3	+37.7	+36.3	+33.7	**+0.0**	**+0.0**	-0.3	+37.3	+37.7	+36.3	**+33.0**	–	+37.3	+33.7
Mouth_Slightly_Open	+13.7	+11.7	+14.0	+5.3	**+0.0**	+0.7	**+0.0**	+13.7	+11.7	+14.0	+5.3	**+0.0**	+12.7	**+0.0**
Mustache	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	–	**+0.0**
Narrow_Eyes	**+0.0**	**+0.0**	**+0.0**	**+0.0**	–	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**
No_Beard	+7.0	+1.7	+5.7	+8.0	**+0.0**	**+0.0**	**+0.0**	+7.0	+1.7	+5.7	+8.0	**+0.0**	+2.3	**+0.0**
Oval_Face	+0.3	**-0.0**	**-0.0**	**-0.0**	**-0.0**	-1.7	**-0.0**	+0.3	**-0.0**	**-0.0**	**-0.0**	**-0.0**	-4.0	**-0.0**
Pale_Skin	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**
Pointy_Nose	-5.0	-7.3	-0.3	+2.0	**+0.0**	**+0.0**	**+0.0**	-5.0	-7.3	-0.3	+2.0	**+0.0**	-2.0	**+0.0**
Receding_Hairline	**+0.0**	**+0.0**	**+0.0**	+0.3	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	+0.3	**+0.0**	**+0.0**	**+0.0**
Rosy_Cheeks	+0.3	**-0.0**	**-0.0**	+1.3	**-0.0**	**-0.0**	**-0.0**	+0.3	**-0.0**	**-0.0**	+1.3	**-0.0**	–	**-0.0**
Sideburns	+0.3	**+0.0**	**+0.0**	+0.7	**+0.0**	**+0.0**	**+0.0**	+0.3	**+0.0**	**+0.0**	+0.7	**+0.0**	-0.3	**+0.0**
Smiling	+30.3	+26.7	+22.3	+16.0	**+0.0**	-0.3	**+0.0**	+30.3	+26.7	+22.3	**+17.3**	–	+25.3	–
Straight_Hair	**+0.0**	+0.3	**+0.0**	+2.3	**+0.0**	**+0.0**	**+0.0**	**+0.0**	+0.3	**+0.0**	+2.3	–	+0.3	**+0.0**
Wavy_Hair	+14.0	+10.7	+8.0	+15.3	**-0.0**	**-0.0**	**-0.0**	+14.0	+10.7	+8.0	+15.3	**-0.0**	–	-0.3
Wearing_Earrings	+2.7	**+0.0**	+0.3	+6.7	**+0.0**	**+0.0**	**+0.0**	+2.7	**+0.0**	+0.3	+6.7	**+0.0**	**+0.0**	**+0.0**
Wearing_Hat	**+0.0**	**+0.0**	**+0.0**	+2.3	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	+2.3	**+0.0**	**+0.0**	**+0.0**
Wearing_Lipstick	+40.0	+38.0	+39.0	+27.7	-4.3	**+0.0**	+0.3	+40.0	+38.0	+39.0	**+30.0**	–	+38.3	+33.0
Wearing_Necklace	-1.0	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	-1.0	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**
Wearing_Necktie	+1.3	**+0.0**	**+0.0**	+2.0	**+0.0**	**+0.0**	**+0.0**	+1.3	**+0.0**	**+0.0**	+2.0	**+0.0**	**+0.0**	**+0.0**
Young	+11.0	+9.3	+11.0	+12.3	**-0.0**	+8.7	**-0.0**	+11.0	+9.3	+11.0	+12.3	**-0.0**	+8.7	+0.3
Mean (pp)	+6.4	+5.1	+5.5	+5.3	-0.4	+0.7	+0.0	+6.4	+4.7	+5.5	+5.3	+0.0	+5.8	+2.6
TanGCE W/T/L				1/38/1							3/31/4			

Table 6: **Per-row λ_i statistics for the proposed pipeline ($\varepsilon = 0.1$, $\lambda_{\max} = 64$) on the 39-panel NLP grid at 50%-depth.** Each cell averages the per-row λ_i first across all 60 outer iterations of one trajectory, then across the 13 model panels of the concept. The within-panel p_{10}/p_{90} spread (mean across panels) is roughly $14/45$, i.e. a $\sim$10$\times$ ratio between the least- and most-confident rows in a single iteration. The safety cap $\lambda_{\max} = 64$ is observed only sporadically (max 57.32 across 39$\times$60 = 2340 iterations); the closed-form rule rarely binds against it.

Concept	n	mean λ_i	median	min	max
Sycophancy	13	29.58	37.15	4.89	49.38
Gender	13	25.28	23.76	0.45	45.08
Safety	13	33.07	33.30	1.71	57.32
Overall	**39**	**29.31**	**30.19**	**0.45**	**57.32**